



NARADA^(R) CARD System

FOR

API SPECIFICATIONS v5.0

Version History

VERSION	AUTHOR	DATE	CHANGE

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1 INTRODUCTION

1.1 Scope

This manual provides detailed API integration guidance to Narada Prepaid Host. Some of the functionality described in this document may be restricted depending on the level of service arrangement with Yalamanchili.

1.2 Prerequisites

- API requires Username and password has to be sent along with clientID [a unique Identifier for the Service Requestor].
- Profile has to be created for the ServiceRequestor in the Narada system. Security Parameters can configure at the Profile Management.
- ServiceRequestor should have a Public-Key pair and share his Public certificate to Yalamanchili.
- Public Certificate of Yalamanchili.
- Service URL (Yalamanchili Hosted).

1.3 Terminology

Narada-API

Narada-API is a secure web services used by ServiceRequestor (Client) to initiate and process the transaction.

ServiceRequestor

The entity using Yalamanchili Prepaid Card System.

2 MESSAGE STRUCTURE AND ELEMENTS

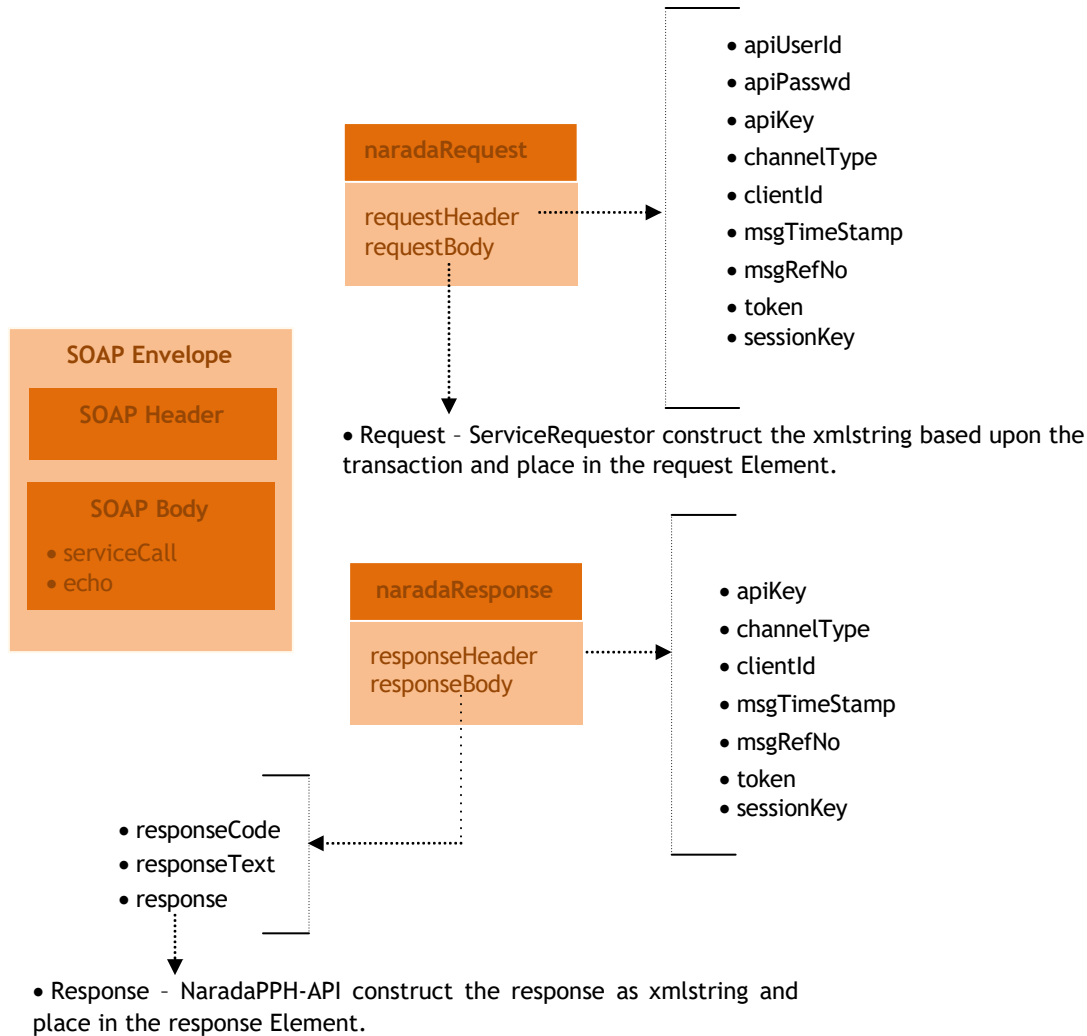
Narada-API Interface is used by ServiceRequestor to avail the PrepaidCard functions using SOAP (Simple Object access protocol).

SOAP is a protocol for exchanging XML-based messages normally using HTTP/HTTPS. The SOAP client (ServiceRequestor) generates the objects and classes based on the Web Services Description Language (WSDL) which is published by Yalamanchili. For further information and references regarding SOAP please refer to the following links:

- <http://en.wikipedia.org/wiki/SOAP>

Elements in Narada-API SOAP

Transaction API



Here ServiceRequestor has to post the transactions to Narada-API System by invoking “serviceCall” Operation. Types of Transactions are identified with key called “apiKey”. Please refer the appendix 5.1 for Transaction List and their apiKeys.

The following is the sample SOAP1.1 request and response of Narada-API.

Request:-

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ser="http://services.ycs.com" xmlns:xsd="http://model.ycs.com/xsd">
<soapenv:Header/>
```

```
<soapenv:Body>
  <ser:serviceCall>
    <ser:naradaRequest>
      <xsd:requestHeader>
        <xsd:apiUserId>String</xsd:apiUserId>
        <xsd:apiPasswd>String</xsd:apiPasswd>
        <xsd:apiKey>String</xsd:apiKey>
        <xsd:channelType>String</xsd:channelType>
        <xsd:clientId>String</xsd:clientId>
        <xsd:msgRefNo>String</xsd:msgRefNo>
        <xsd:msgTimeStamp>String</xsd:msgTimeStamp>
        <xsd:sessionKey>String</xsd:sessionKey>
        <xsd:token>String</xsd:token>
      </xsd:requestHeader>
      <xsd:requestBody>
        <xsd:request>xml as string</xsd:request>
      </xsd:requestBody>
    </ser:naradaRequest>
  </ser:serviceCall>
</soapenv:Body>
</soapenv:Envelope>
```

Refer to the [section 2.1](#) for detailed narrative of RequestHeader and RequestBody Elements.

Response:-

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ser="http://services.ycs.com" xmlns:xsd="http://model.ycs.com/xsd">
  <soapenv:Header/>
  <soapenv:Body>
    <ser:serviceCallResponse>
      <ser:naradaResponse>
        <xsd:responseHeader>
          <xsd:apiKey>String</xsd:apiKey>
          <xsd:channelType>String</xsd:channelType>
          <xsd:clientId>String</xsd:clientId>
          <xsd:msgRefNo>String</xsd:msgRefNo>
          <xsd:msgTimeStamp>String</xsd:msgTimeStamp>
          <xsd:sessionKey>String</xsd:sessionKey>
          <xsd:token>String</xsd:token>
        </xsd:responseHeader>
        <xsd:responseBody>
          <xsd:responseCode>String</xsd:responseCode>
          <xsd:responseText>String</xsd:responseText>
          <xsd:response>xml as string</xsd:response>
        </xsd:responseBody>
      </ser:naradaResponse>
    </ser:serviceCallResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Refer to the [section 2.1](#) for detailed narrative of ResponseHeader and ResponseBody Elements.

2.1 Transaction Request and Response Elements

This section explains the fields which are need to be passed in the request and response of the transaction. Except apiUserId and apiPasswd fields others will be echoed in the response header.

Request Elements

The following table represents the elements required for submitting a transaction request to the Narada-API

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
requestHeader			
apiUserId	String {8}	Yes	User Name for Authentication. This is a fixed value of 8 characters Client Team has to ensure to generate a 8 character USERID for API
apiPasswd	String	Yes	Password for Authentication. YALAMANCHILI Team would submit this value for a given USER-ID. To Hash the password with SHA-256 and encode to BASE64 format and eliminate last character.
apiKey	String {6}	Yes	This field is used to identify the type of Transaction. Refer the Appendix for apiKey mapped to Transaction. This is a fixed length field.
channelType	String {6}	Yes	This filed value describes the request origination point.Ex:- IVR/WEB/ATM/KIOSK/API The field is a variable field and can be send without any padding to the length, however maximum length is 6.
clientId	String {6}	Yes	Client Identifier generated by Yalamanchili System
msgRefNo	String {12}	Yes	This tag will hold a unique reference number that"s generated by the ServiceRequestor which will be used to track each and every request. If value is lesser than 12 bytes right pad with numeric „0" to make it to 12 bytes. For Ex : 100000000000
msgTimeStamp	String {14}	Yes	Time stamp of the Request Origination in the format of YYYYMMDDHHMISS

			The timezone has to be the client country timezone.
sessionKey	String	Conditional	If encryption is enabled, the given symmetric key for Decryption. If encryption is not enabled empty tag is still required
Token	String	Conditional	If encryption is enabled, the given Signed Data for Verify If encryption is not enabled empty tag is still required
requestBody			
request	String	Yes	XML String of Transaction"s API parameters. Refer the Transaction API for Parameter"s to be constructed as XML String.

Response Elements

The following table lists the fields returned in the response from the Narada API.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
responseHeader			
apiKey	String {6}	Yes	Echoed back from Request
channelType	String {6}	Yes	Echoed back from Request
clientId	String {6}	Yes	Echoed back from Request
msgRefNo	String {12}	Yes	Echoed back from Request
msgTimeStamp	String {14}	Yes	Time stamp of the Response Origination in the format of YYYYMMDDHHMISS
sessionKey	String	Optional	Symmetric key for Decryption
Token	String	Optional	Signed Data
responseBody			
responseCode	String {6}	Mandatory	Indicates Successful and non successful Response.
responseText	String	Mandatory	Response Code Description
response	String	Mandatory	Transaction Response formatted as XML String. Refer the API list for parameters in the xml string.

Note: response field value will be set only when the response code ("00") is success.

HeartBeat API

Here ServiceRequestor has to post the heart beat request to "NaradaPPH-API" System by invoking "echo" Operation. By this requester would know whether service is alive or not.

The following is the sample SOAP1.1 request and response of echo.

Request:-

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ser="http://services.api.ycs.com">
  <soapenv:Header/>
  <soapenv:Body>
    <ser:echo>
      <ser:echoReq>Test, Connection</ser:echoReq>
    </ser:echo>
  </soapenv:Body>
</soapenv:Envelope>
```

Refer to the [section 2.2](#) for detailed narrative of RequestHeader and RequestBody Elements.

Response:-

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  <soapenv:Body>
    <ns2:echoResponse xmlns:ns2="http://services.api.ycs.com">
      <ns2:echoResp>Welcome To Prepaid API, Request received from client : [Test,
Connection]</ns2:echoResp>
    </ns2:echoResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

2.2 HeartBeat Request and Response Elements

This section explains the fields which are need to be passed in the request and response of the heart beat. It will be echoed in the response.

Request Elements

The following table represents the elements required for submitting a heart beat request to the NaradaPPH-API

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
echo			
echoReq	String	Yes	Any text message

Response Elements

The following table lists the fields returned in the response from the NaradaPPH echo API.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
echoResponse			
echoResp	String	Yes	Echoed back from Request

3 INTEGRATION APPROACH

- Have the WSDL URL or static WSDL file along with Service End Point URL from Yalamanchili.
- APIUserId and Password for Authentication. APIUserId should be created in the Narada Admin Portal under User Access Management and then assign the access rights for the User.
- Share the Public certificate between both entity (Yalamanchili and ServiceRequestor) for message encryption. Refer the Section “Security Implementation for in detail”. Message encryption is optional if the both entity communicates over MPLS/VPN over SSL.
- RequestHeader and RequestBody all elements should be constructed in the same order as that on WSDL file.
- Parameter Specification document for the Transactions supported. Refer the section-4 “Submitting a Transaction”
- Once the above information has gathered, then start integrating with Narada-API Interface.
- The Java sample pseudo codes (created using Eclipse IDE with Axis2 runtime) provided below describe how to invoke each of exposed methods. These code samples only include the mandatory fields to be provided by the ServiceRequestor.
- Add a service reference to provided URL / create a proxy using the WSDL and generates the stubs and classes. (Here APIServiceStub is created along with other classes)
- The API should support operation style is document/literal.
- SOAPUI client simulator tool will not do this process for our request xml data. So it should be replaced manually for request XML data and that should be given in the SOAPUI request. This replacing is required only while testing through SOAPUI. For testing through proper client code, it doesn't require.
 - Xml starting character „<“ to be replaced with <
 - Xml ending character „>“ to be replaced to >

For Ex: <RequestXml> <TxnRefNo>12345678</RequestXml> </RequestXml>

Note: For Production End point URL /Public Certificate/ API UserId /API Password will be different.

```
NaradaApiServiceStub serviceStub = new NaradaApiServiceStub
("http://localhost:5050//prepaid/service/APIService");

//HeartBeat API Request Samples
//-----
Echo echo = new Echo();
echo.setEchoReq("Test, Connection"); //HeartBeat API request.
EchoResponse er = serviceStub.echo(echo);
System.out.println("HeartBeat API Response : "+er.getEchoResp() ); //HeartBeat API Response

//Transaction API Request samples
//-----
ServiceCall service = new ServiceCall();
NaradaRequest request = service.getNaradaRequest();

RequestHeader header = new RequestHeader();
header.setApiUserId("TEST01");
header.setApiPasswd("secret");
header.setApiKey ("APNCRD"); // Refer the Appendix for APIkey for various transaction.
header.setChannelType("API");
```

```

header.setClientId("CLTN01");
header.setMsgRefNo("123456123456");
header.setMsgTimeStamp ("20130425022520");
header.setSessionKey(""); //Optional. This field is required when sending message as encrypted.
header.setToken(""); //Optional. This field is required when sending message as encrypted

RequestBody body = new RequestBody();
/*Construct the request XML and place in the request field. Encode the XML entity characters for the
constructed xml if SOAP framework doesn't do. There should not be any spaces between the xml
elements
*/
/* Sample xml build for Register the Customer */
String requestxml="
<RequestXml><TxnRefNo>100000000001</TxnRefNo><ProxyNumber>9999000000405</ProxyNumber><Cust
omerInfo><Title>Mr</Title><FirstName>Karthick</FirstName><LastName>Kannan</LastName><DOB>0201
1980</DOB><Gender>M</Gender><MobileNumber>+919884012345</MobileNumber><EmailAddress>test@y
ala.com</EmailAddress><Address><Address1>No 11/6</Address1><Address1>Balaji Avenue 1st
Street</Address1><Address2>T.Nagar</Address2><City>Chennai</City><State>TamilNadu</State><Count
ry>Chennai</Country><PostalCode>600017</PostalCode></Address><MailAddress><Address1>#143 Sairam
Colony</Address1><Address2>BrindavanNagar</Address2><City>Chennai</City><State>TamilNadu</State
><Country>Chennai</Country><PostalCode>600012</PostalCode></MailAddress><IdentificationType><ID
Type>Passport</IDType><IDNumber>ABCD1234TF</IDNumber><ExpiryDate>09092020</ExpiryDate><Coun
tryofIssue>India<CountryofIssue></IdentificationType>/CustomerInfo><LoadDetails><ProductCode>PRD00
1</ProductCode><LoadAmount>100</LoadAmount><BillingAmount>100</BillingAmount><LoadCurrency>I
NR</LoadCurrency><BillingCurrency>INR</BillingCurrency></LoadDetails></RequestXml>  ";

body.setRequest(requestXml);

request.setRequestHeader(header);
request.setRequestBody(body);

service.setNaradaRequest(request);

ServiceCallResponse serviceResponse = serviceStub.serviceCall(service);

NaradaResponse response = serviceResponse.getNaradaResponse();
ResponseBody resBody = response.getResponseBody();
ResponseHeader resHeader = response.getResponseHeader();

String apiKey= resHeader.getApiKey();
String responseCode = resBody.getResponseCode();
String responseText = resBody.getResponseText();
String responseBody = resBody.getResponse(); //Decode the XML entity characters if required.

```

4 SUBMITTING TRANSACTIONS

4.1 Purchase a Card/Register the customer - APNCRD

This API is used to purchase a personalized prepaid card or Register the Instant Card (Non-personalized) or virtual card with the System.

The Purchase transaction involves registration of the customer, product assignment and create card Number along with proxy Number. Proxy Number is pseudo Card Number and it will be used as a proxy to prevent the exposure of card Number to external world.

Based on Business requirements cards can be activate instantly or activate later using the activation code through any of the channels and are Narada-API/ Customer Portal/ Business office Portal/ IVR. To activate via Narada-API use the “Card Activation API- APCDAC”.

New card Request

While initiating a New Card transaction, then following parameters has to be sent to the Prepaid API System.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ProxyNumber	String{16}	Optional	Mandatory for Instant Card Registration.
CustomerId	String {100}	Optional	Customer Identifier generated by ServiceRequestor and used to map the customer of ServiceRequestor proprietary System with Yalamanchili Prepaid System.
NameOnCard	String {25}	Optional	Customer Preferred Name on the Card. This filed is applicable for Purchasing a personalized Card.
CardDesign	String {6}	Optional	This field can be used to mention the Plastic Design required for the card. This is used for the personalized card issuance. If the Value not Present then Default Design will be in use.
CardActivationFlag	String {1}	Optional	Set to “Y” if cards need to be activate immediately and “N” to activate later. This field is applicable only for the Instant Card and Virtual Card. If the value is “N” then the activation code is generated and available in the New Card Response. This activation code is used to activate the card through the Card Holder Portal /IVR
AlertFlag	String {1}	Optional	For Future Use.
RefText1	String {32}	Optional	Used to store the value in 4 th line embossing filed. And same can be provided in embossing file to card Bureau

RefText2	String {6}	Optional	Used to send the campaign code The campaign code is a separate feecode that is maintained in the system through the NARADA(R) Fee Management Functionality function in the front-end portal. This is usually a 6 digit code that is used to assign specific fee settings for a given card which will override the Fee Settings that is mapped to the Product. A typical example where this is used is to maintain a differential charging for VIP Customers.
RefText3	String {32}	Optional	For Future Use
RefText4	String {32}	Optional	For Future Use
RequestXml/CustomerInfo			
Title	String {5}	Yes	Customer"s Title
FirstName	String {25}	Yes	Customer"s First Name
LastName	String {25}	Yes	Customer"s Last Name
DOB	String {8}	Yes	Customer"s Date of Birth. Format : DDMMYYYY
Gender	String {1}	Yes	Customer"s Gender [M/F]
MobileNumber	String {15}	Yes	Customer"s Mobile Number Format: (+) with Country Code.
EmailAddress	String{50}	Yes	Customer"s Email Address
Image	String	No	This field is applicable for Personalized Card. Used to upload the Customer"s Photo. Format: Encode the byte to Base64 Format.
RequestXml/CustomerInfo/Address			
Address1	String {35}	Yes	Customer"s residence address Line No 1
Address2	String {35}	Yes	Customer"s residence address Line No 2
Address3	String {35}	Optional	Customer"s residence address Line No 3
Address4	String {35}	Optional	Customer"s residence address Line No 4
City	String {20}	Yes	Customer"s residence address City Name
State	String {20}	Yes	Customer"s residence address State Name
Country	String {20}	Yes	Customer"s residence address Country Name
PostalCode	String {15}	Yes	Customer"s residence address Postal Code
HomeNumber	String {15}	Optional	Customer"s contact Number
RequestXml/CustomerInfo/MailAddress			
Address1	String {35}	Yes	Customer"s business address Line No 1

Address2	String {35}	Yes	Customer"s business address Line No 2
Address3	String {35}	Optional	Customer"s business address Line No 3
Address4	String {35}	Optional	Customer"s business address Line No 4
City	String {20}	Yes	Customer"s business address City Name
State	String {20}	Yes	Customer"s business address State Name
Country	String {20}	Yes	Customer"s business address Country Name
PostalCode	String {15}	Yes	Customer"s business address Postal Code
WorkNumber	String {15}	Optional	Customer"s business Work Telephone Number
Extension	String {5}	Optional	Customer"s business Work Number Extension
RequestXml/CustomerInfo /IdentificationType - This node can be multiple when submitting more than one proof and it is optional tag			
IDType	String {25}	Yes	Customer"s Identification Proof Type
IDNumber	String {20}	Yes	Customer"s Identification Proof Type Number
ExpiryDate	String {8}	Optional	Expiry Date of Customer"s Identification Proof Format :DDMMYYYY
CountryofIssue	String {20}	Yes	Identification proof issued county Name.
RequestXml/LoadDetails			
ProductCode	String {6}	Yes	Short Code of product to map with the Card.
LoadAmount	String {12}	Yes	Amount to be loaded into the Card account of the card holder
BillingAmount	String {12}	Yes	This field contains the amount in the card account currency
LoadCurrency	String {3}	Yes	Currency used by the Card holder to fund the card account. For example, if Card holder used USD to load a GBP Card account, then Loading currency will be USD
BillingCurrency	String {3}	Yes	This is the currency of the card account

New card Response

The following table lists the fields returned in the response from the Narada-PPH API along with ResponseCode and ResponseText. Refer the Appendix for the Code and Information.

For Virtual card /Card Not Present Product ProxyNumber will not be created. Now the Card is created and respective product will be linked .And Card Status will be inactive status which allows loading the amount to account.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	For Personalized card it will be generated and set to the field. If it is Instant card then echoed back the Proxy Number.

CardNumber	String{16}	Optional	CardNumber will be present only for Virtual Card/Card Not Present Products.
CustomerId	String {100}	Optional	Echoed back from the Request
ActivationCode	String { 6 }	Optional	6 digits Number will be set. As default Activation Code will be generated for Personalized Card. For Other Cards it will be based on Card Activation Flag.
CVV	String{3}	Optional	CVV2 value will be set in the Response only for the Virtual Card.
CardExpiry	String{8}	Optional	Expiry Date Value will be set to the response only for the Virtual Card.

4.2 Update Customer Details - APUPCR

Update Customer Request -

This API can be called to update the details of an existing Customer/Card holder. The details those are allowed for an update will be the email ID, mobile number and contact addresses. Customer Name cannot be updated.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
RequestXml/CustomerInfo			
Title	String {5}	Yes	Customer"s Title
FirstName	String {25}	Yes	Customer"s First Name
LastName	String {25}	Yes	Customer"s Last Name
DOB	String {8}	Yes	Customer"s Date of Birth. Format : DDMMYYYY
MobileNumber	String {15}	Yes	Customer"s Mobile Number. Format : Prefix with (+) and along with Country Code Ex: +6598804000
EmailAddress	String{50}	Yes	Customer"s Email Address
RequestXml/CustomerInfo/Address - This Node will be Mandate if need to update Customer Residence Address			
Address1	String {35}	Yes	Customer"s residence address Line No 1
Address2	String {35}	Yes	Customer"s residence address Line No 2
Address3	String {35}	Optional	Customer"s residence address Line No 3
Address4	String {35}	Optional	Customer"s residence address Line No 4
City	String {20}	Yes	Customer"s residence address City Name
State	String {20}	Yes	Customer"s residence address State Name
Country	String {20}	Yes	Customer"s residence address Country Name
PostalCode	String {15}	Yes	Customer"s residence address Postal Code

HomeNumber	String {15}	Optional	Customer"s contact Number
RequestXml/CustomerInfo/MailAddress - This Node will be Mandate if need to update Customer Mailing Address			
Address1	String {35}	Yes	Customer"s business address Line No 1
Address2	String {35}	Yes	Customer"s business address Line No 2
Address3	String {35}	Optional	Customer"s business address Line No 3
Address4	String {35}	Optional	Customer"s business address Line No 4
City	String {20}	Yes	Customer"s business address City Name
State	String {20}	Yes	Customer"s business address State Name
Country	String {20}	Yes	Customer"s business address Country Name
PostalCode	String {15}	Yes	Customer"s business address Postal Code
WorkNumber	String {15}	Optional	Customer"s business Work Number
Extension	String {5}	Optional	Customer"s business Work Number Extension
RequestXml/CustomerInfo/IdentificationType - This node can be multiple when submitting more than one proof and it is optional tag			
IDType	String {25}	Yes	Customer"s Identification Proof Type
IDNumber	String {20}	Yes	Customer"s Identification Proof Type Number
IssueDate	String {8}	Optional	Customer"s Identification Proof issue Date Format : DDMMYYYY
ExpiryDate	String {8}	Optional	Expiry Date of Customer"s Identification Proof Format :DDMMYYYY
CountryofIssue	String {20}	Yes	Identification proof issued county Name.
IDImage	String	Optional	It is used to upload the identification image. Format: Encode the byte to Base64 Format. More than one images can be uploaded by giving comma separator Example : Image1,Image2
IDImageName	String	Optional	It is used to upload the identification image name with any one of the following extension Image format : JPG, GIF, PNG and BMP More than one image name can be uploaded by giving comma separator Example : Image1.JPG, Image2.JPG
RequestXml/CustomerInfo/FutureUse - This Node will be future use.			
Reftext1	String	Optional	It will be return the xml format and always should be same format.

			For Ex : <OtherDetails> <Empld>XXX</Empld> <EmplssueDt>DDMMYYYY</EmplssueDt> <EmpExpiryDt>DDMMYYYY</EmpExpiryDt> <EmiratesId>XXX</EmiratesId> <EmiratesCardNo>XXX</EmiratesCardNo> <EmiratesIssueDt>XXX</EmiratesIssueDt> </OtherDetails>
Reftext2	String	Optional	Future Use2
Reftext3	String	Optional	Future Use3

Update Customer Response

Once the data is updated successfully in the system, then you will get the responseXML or else error will be passed in response code field, the following table lists the fields returned in the response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Mandatory	Echoed back from the Request
ProxyNumer	String {16}	Mandatory	Echoed back from the Request

4.3 Get Card Details - APCHDT

This API can be called to retrieve the account summary of the card holder. This will return the card holder details, lists of accounts and the linked to the card number.

Get Card Details Request

Customer"s CardNumber is the primary field to fetch the card status, products linked to the card, amount on the account and customer details.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor
ProxyNumber	String {16}	Yes	Either Card Number or Proxy Number can be accepted. CardNumber will be 16 digits and ProxyNumber will be 12 digits.
CustomerId	String {100}	Optional	It is used to map the customer of ServiceRequestor proprietary System with Yalamanchili Prepaid System An example is the Customer ID of the Financial

			Institution Core Banking system for the particular customer who is given a Payment Card
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Get Card Details Response

In the response ServiceRequestor can able to get card information like status of the card as the form of short code. (Refer the Appendix for the card status), customer details and linked account details. If the customer has multiple products mapped to single card, then those details will be presented in the responseXML as multiple based on number of products.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.
CardStatus	String {2}	Yes	Status of the Card
ExpiryDate	String {8}	Yes	Expiry Date of the Card. Format : DDMMYYYY
NameOnCard	String {25}	Optional	Customer Preferred Name on the Card. This filed is applicable for Purchasing a personalized Card.
ResponseXml/CustomerInfo			
Title	String {5}	Optional	Customer"s Title
FirstName	String {25}	Optional	Customer"s First Name
MiddleName	String {25}	Optional	Customer"s Middle Name
LastName	String {25}	Optional	Customer"s Last Name
DOB	String {8}	Optional	Customer"s Date of Birth. Format : DDMMYYYY
Gender	String {1}	Optional	Customer"s Gender [M/F]
MobileNumber	String {15}	Optional	Customer"s Mobile Number
EmailAddress	String {50}	Optional	Customer"s Email Address
Nationality	String {20}	Optional	Customer"s Nationality
MotherName	String {25}	Optional	Customer"s Mother Name
RequestXml/CustomerInfo/Address - This Node will be Mandate if need to Customer Residence Address			
Address1	String {35}	Optional	Customer"s residence address Line No 1
Address2	String {35}	Optional	Customer"s residence address Line No 2
Address3	String {35}	Optional	Customer"s residence address Line No 3
Address4	String {35}	Optional	Customer"s residence address Line No 4
City	String {20}	Optional	Customer"s residence address City Name
State	String {20}	Optional	Customer"s residence address State Name
Country	String {20}	Optional	Customer"s residence address Country Name
PostalCode	String {15}	Optional	Customer"s residence address Postal Code
HomeNumber	String {15}	Optional	Customer"s contact Number
RequestXml/CustomerInfo/MailAddress - This Node will be Mandate if need to Customer Mailing Address			

Address1	String {35}	Optional	Customer" s business address Line No 1
Address2	String {35}	Optional	Customer" s business address Line No 2
Address3	String {35}	Optional	Customer" s business address Line No 3
Address4	String {35}	Optional	Customer" s business address Line No 4
City	String {20}	Optional	Customer" s business address City Name
State	String {20}	Optional	Customer" s business address State Name
Country	String {20}	Optional	Customer" s business address Country Name
PostalCode	String {15}	Optional	Customer" s business address Postal Code
WorkNumber	String {15}	Optional	Customer" s business Work Number
Extension	String {5}	Optional	Customer" s business Work Number Extension
RequestXml/CustomerInfo/FutureUse - This Node will be future use.			
Reftext1	String	Optional	It will be return the xml format and always should be same format. For Ex : <OtherDetails> <EmpId>XXX</EmpId> <EmplIssueDt>DDMMYYYY</EmplIssueDt> <EmpExpiryDt>DDMMYYYY</EmpExpiryDt> <EmiratesId>XXX</EmiratesId> <EmiratesCardNo>XXX</EmiratesCardNo> <EmiratesIssueDt>XXX</EmiratesIssueDt> </OtherDetails>
Reftext2	String	Optional	It will be return the OFAC card status.
Reftext3	String	Optional	Future Use3
RequestXml/CustomerInfo/IdentificationType - This node can be multiple when submitting more than one proof and it is optional tag			
IDType	String {25}	Optional	Customer" s Identification Proof Type
IDNumber	String {20}	Optional	Customer" s Identification Proof Type Number
IssueDate	String{8}	Optional	Customer" s Identification Proof issue Date Format : DDMMYYYY
ExpiryDate	String {8}	Optional	Expiry Date of Customer" s Identification Proof Format :DDMMYYYY
CountryofIssue	String {20}	Optional	Identification proof issued county Name.
IDImage	String	Optional	It is used to upload the identification image. Format: Encode the byte to Base64 Format. More than one images can be uploaded by giving comma separator

			Example : Image1,Image2
IDImageName	String	Optional	It is used to upload the identification image name with any one of the following extension Image format : JPG, GIF, PNG and BMP More than one image name can be uploaded by giving comma separator Example : Image1.JPG, Image2.JPG
ResponseXml/ProductInfo/Details			
Amount	String{12}	Optional	Balance of the wallet
ProductCode	String{6}	Optional	Product code of the Wallet
Currency	String{3}	Optional	ISO Currency code of the Balance Amount
CampaignCode	String{6}	Optional	Campaign code of the Wallet

4.4 Block / Lock / Unlock the Card - ACDSTS

This API is used to change the status of the card. For example when the Consumer Card's got lost or damaged or stolen then he has to block the card to avoid Fraud transaction. Various statuses can be marked to the Card in the System based on the reason to block.

- BL - Permanently Block the Card due to card Lost
- BS - Permanently Block the card due to Card Stolen
- BD - Permanently Block the Card due to physical Damage
- L - Lock the Card and this can be unlocked by giving „UL“ flag to this API

Once the card is blocked, then online transactions (ATM, POS, and ECOM) are not allowed. But web transactions can be allowed i.e. Card Holder can use the portal to view his transaction and we can't do any of the below activity for temporarily blocked card.

- Card load
- Account addition
- Fund Transfer
- Account closure
- Refund
- Debit/credit Account

Locked card can be unlocked, Card Holder can't place a replacement card Request for locked card. Permanently blocked cards can't be unblocked (BL/BS/BD) only card replacement can be done. To request Replacement Card via Narada-API call "Card Replacement API-APCDRP".

Locked card can be unlocked by sending the flag parameter as "UL" - Unlock the Card. Once the card unlocked then Card Holder can perform Online Transactions.

S.No	New Status to be changed	Allowed status of card
1	Card Locked (L)	Active
2	Card Blocked BL : lost BS : stolen BD : damaged	Active
3	Card Unlocked (UL)	Locked

Card Block/Unblock Request

The following table lists the fields which need to send in the Request.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
flag	String{2}	Yes	Status of the card needs to be updated. Values are BL/BS/BD/L/UL
Reason	String {25}	Yes	Reason for Block/Lock/unlock.

Card Block/unblock Response

If the card is already in requested status, then system will not change the card status and send the non successful response Code and with the description.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	Echoed back from the Request

4.5 Card Activation - APCDAC

Once the Card is created successfully through the “Purchase a Card API” and you will get a ProxyNumber, Card Number along with 6 digit activation code. Use the activation code along with ProxyNumber to activate the card.

Card Activation Request

The following table lists the fields which need to send in the Request.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
ActivationCode	String {6}	Optional	Activation Code to activate the card.
CustomerId	String {100}	Optional	It is used to map the customer of ServiceRequestor proprietary System with Yalamanchili Prepaid System An example is the Customer ID of the Financial Institution Core Banking system for the particular customer who is given a Payment Card

Card Activation Response

The following table lists the fields which returned in the Response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	Echoed back from the Request

If you want to know the status of card and balance use the “Get Card Details API - APCHDT”

4.6 Card Replacement - APCDRP

This API is used to place a replacement request for two types of cards

1. Replace the existing Physical Card.
2. Get Physical Card for Virtual Card with same Number.

When to Replace the existing Physical Card:-

The existing card should be blocked before applying a replacement card. Replacement process involved the creation of card, Pin mailer, Card Embossing and Card Delivery and those functions will be at the End of Day Processing. Card Replacement will be applicable based on upon the product configuration in the Narada System.

Virtual to Physical Card:-

When the customer wants to convert his virtual card to physical Card with the different Virtual card Number as physical, then use the type as “VP” - Virtual to Physical in the request parameter.

Card Replacement Request

The following table list the fields which are need to send in the request. Type is the keyword parameter to identify the replacement type.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
NewProxyNumber	String {16}	Conditional	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits. This is mandatory in case of non-personalized/instant cards
CustomerId	String {100}	Optional	Customer Identifier (Unique Registration Number - URN) generated by ServiceRequestor and used to map the customer of ServiceRequestor proprietary System with Yalamanchili Prepaid System.
Type	String {2}	Yes	VP - Virtual to Physical PP - Physical to Physical.
CardDesign	String {6}	Optional	To specify the card design for the replacement. If the value is not set then default design will be taken.
DeliveryFlag	String {1}	Conditional	Based on the delivery flag card will be delivered B - Branch Address C- Customer Address It is mandatory for personalized replacement cards only

Card Replacement Response

On successful transaction, replacement card request is placed in the system and by End of Day processing card Number will get generated and once Card is created then it will be delivered to Consumer/Client based on the mailing information.

If the card is not blocked already, then system will send non successful Response Code. Note: If you receive the non-successful response and then responseXml will not be set in the response field. The fee will be charged for the customer if it is configured for the transaction. In addition to that if the card design is not mapped for the product or Inputted Design not available for the product then request will not be processed successful.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			

TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	Echoed back from the Request
NewProxyNumber	String {16}	Optional	NewProxyNumber will be present only for Virtual Card
CVV	String {3}	Optional	CVV2 value will be set in the response only for the Virtual Card.
ExpiryDate	String {8}	Optional	Expiry Date will be set in the response only for the Virtual Card.

4.7 Account Addition - APACAD

Narada supports Consumer to have more than one product in a Single Card. This API is can be used whenever customer want to subscribe a new product to his card.

Account Addition Request

The following table list the fields which are need to be sent in the request. If ServiceRequestor wants to add more than one product in same request, then he has to construct details element node for each product under the ProductInfo Node.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
RequestXml/ProductInfo/Details			
ProductCode	String {6}	Yes	Short Code of product to subscribe.
LoadAmount	String {12}	Yes	Amount to be loaded into the Card account of the card holder with amount will be zero or greater than zero
BillingAmount	String {12}	Yes	This field contains the amount in the card account currency
LoadCurrency	String {3}	Yes	Currency used by the Card holder to fund the card account. For example, if Card holder used USD to load a GBP Card account, then Loading currency will be USD Format: 3 digits ISO Currency Code.
BillingCurrency	String {3}	Yes	This is the currency of the card account. Format: 3 digits ISO Currency Code.

Account Addition Response

On the successful transaction, new product is added to the Card and now Card Holder can start using his new product. If the product is already available for the card, then system will be sent non successful response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	Echoed back from the Request

4.8 Load Card / Funding an Account - APLCRD

This API is used to load the funds in to customer account and when card is active and inactive (It is possible status before card gets activated).

S.No	Load Card	Allowed status of card
1	Load the funds	Active/ Inactive (It is possible status before card gets activated)
2	Fund cannot be load	Closed/Locked/Unlocked/Blocked (Lost, Stolen, Damaged)

Load Card Request

The following table list the fields which are need to be sent in request.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
RequestXml/ProductInfo			
ProductCode	String {6}	Yes	Short Code of product
LoadAmount	String {12}	Yes	Amount to be loaded into the Card account of the card holder
BillingAmount	String {12}	Yes	This field contains the amount in the card account currency
LoadCurrency	String {3}	Yes	Currency used by the Card holder to fund the card account. For example, if Card holder used USD to load a GBP Card account, then Loading currency will be USD. Format: 3 digits ISO Currency Code.
BillingCurrency	String {3}	Yes	This is the currency of the card account. Format: 3 digits ISO Currency Code.

Load Card Response

Card account balance will be updated online on the Successful execution of transaction .If any fees have been configured in the Product Definition for load transaction, it will get deducted and the

remaining amount will be available as balance amount of the Card account. The following table lists the fields which are returned in the response.

The scenario where transaction get fails is

- While trying for the closed card
- If Card Balance exceeds the Balance limit
- Transaction limit validation (daily, weekly, monthly) based on Limit rules configure for the Product.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request
ProxyNumber	String{16}	Yes	Echoed back from the request
BalanceAmt	String{12}	Yes	Available balance on the account
FeeAmount	String{12}	Yes	Shows the fee amount if any charged or else value will be zero.

4.9 Query Balance - APACBL

This is API is used to get the balance of the card.

Query Balance Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.

Query Balance Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String{16}	Yes	Echoed back from the Request
ProductInfo/Details			
ProductCode	String{6}	Yes	Short code of the product.
BalanceAmount	String{12}	Yes	Balance Amount on the card.
AccountCcy	String{3}	Yes	Card Account Currency. Format: 3 digits ISO Currency Code.

4.10 Amount Refund - APCRND

This API can be called to place a request for an amount to refund from a card account. Two Type of refund will be supported are Partial and Full Refund. If fees has defined in the product definition for refund transaction, then balance of the card = Refund amount - Fee. If the Refund amount + fee Amount is greater than balance then refund request will get rejected.

Refund Request

The following table list the fields which are need to send in the request.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String{6}	Yes	Product from where amount to be refunded
Amount	String {12}	Yes	Refund Amount. For full refund set the amount as "***" in the field. And for partial specify the exact amount to refund

Refund Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request
ProxyNumber	String{16}	Yes	Echoed back from the request
ProductCode	String{6}	Yes	Echoed back from the request
BalanceAmt	String{12}	Yes	Available balance (Refund amount - fee).
FeeAmount	String{12}	Yes	Shows the fee amount if any charged or else value will be zero.

4.11 Account closure - APACCL

This API can be used to close the account (Product) associated with the card based on the product code in the request parameter. If the Card linked to more than one product, then based on product send in the request alone get closed other accounts will remains in active and card also in active status.

In the scenario when the card has single product and tried for account closure, then below action are performed in system are

- Check whether the balance on the account is zero. If matched then the account will be closed.
- Check whether any other products linked to card. If no then close the card too.

Note: Account will be eligible to close when the balance is zero.

Account Closure Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String{6}	Yes	Product from where amount to be refunded

Account Closure Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String{16}	Yes	Echoed back from the request.
ProductCode	String {6}	Yes	Echoed back from the request

4.12 Get X Transactions - APXTXN

This API is used to fetch the transactions performed on the card. The count of the transactions required in the response should be set in field “count” as part of input parameter

If the input value for count parameter is provided as „***” then irrespective of the „From”, „To” dates, all the available transactions are set in the response (Maximum transaction returned in the response is “99”).

Number of records per page can be specified in the parameter 'COUNT'. The application logic would be that if PageNumber XML Tag is filled then we will assume that COUNT contains the Number of records per page.

Get X Transaction Request

FromDate Value should not be greater than six months from current date. And also difference between FromDate and ToDate will be three 90 days.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.

ProxyNumber	String{16}	Yes	Customer"s Card Number.
FromDate	String{8}	Yes	From Date of the Transaction. Format : DDMMYYYY
ToDate	String{8}	Yes	End Date of the Transaction. Format : DDMMYYYY
Count	String{2}	Yes	Number of transaction to be returned, the count should be <= 99. If value is set to "***" then by default maximum number of transaction available in response is "99".
PageNumber	String{2}	Optional	if the transaction count is > 99, records will be divided into pages based on the given count, and corresponding Page number to be given (e.g assume the total number of records is 200, Count value is given as 50 and page number is given as 3 means, the response will display the records from 101 to 150. i.e 50 records per page)

Get X Transaction Response

Transaction response will not be irrespective to the account/product of the card. For example if two products are linked to the card, then transaction will be fetched between the request date will be collection of transaction done on the two products.

This API will return settled, unsettled and declined transaction irrespective of transaction status for the given date range. Fee details will be returned in separate row along with the original transaction, Transaction reference number or auth id will be the identification field to group the original transaction and its fee transaction

Following are the total fields applicable for transaction response as part of common API specification. Based on client requirement and usage, we can enable or disable the fields.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Echoed back from the request.
ProxyNumber	String{16}	Yes	Echoed back from the request.
ActivateDate	String{8}	Yes	Date when card was activated. Format DDMMYYYY
Transactions/Detail			
TxnRefNo	String{32}	Yes	Transaction reference number - 16 digits for web transaction, 32 digits for online/API transaction (APCRND,ACDSTS,APCRLD,APFNTR,APRND)
TransactionDate	String{11}	Yes	The date that the transaction was processed (DD-MMM-YYYY)
TransactionTime	String{8}	Yes	The time that the transaction was process (HH:MI:SS)
ProductCode	String{6}	Yes	Card"s product code (e.g. DBPRD1,DBPRD2)
TransactionCode	String{6}	Yes	Yalamanchili"s Internal transaction code, (e.g

			010000 - ATM Cash withdrawal, 000000 - POS Purchase)
TransSettleDate	String{8}	Optional	Transaction settlement date, it will be null for unsettled transaction
TransRefNo	String{50}	Yes	Unique transaction ID
Description	String{50}	Yes	Transaction description (e.g. Purchase, ATM cash withdrawal, International cross currency fee)
TransCurrency	String{3}	Yes	Transaction Currency. Format: 3 digits ISO Currency Code.
TransStatus	String{100}	Yes	Transaction Status like Pending, settled, Chargeback raised, copy request raised, Outstanding marked, dispute settled, Transaction reversed and etc.
TransAmt	String{12}	Yes	Transaction Amount
BillingCurrency	String{3}	Yes	Currency of the card Amount. Format: 3 digits ISO Currency Code.
BillingAmt	String{12}	Yes	Amount in account currency. The value is present if the amount debited / credited to the account.
DrCrIndicator	String{1}	Yes	"D" - Account Debit transaction, "C" - Account Credit transaction
ExRate	String{12}	Yes	The exchange rate as applied to the transaction should be available. If the transaction is happening in the card currency, then the exchange rate should be defaulted to 1.0000
AuthId	String{6}	Optional	Authorisation Code of the transaction done. Will be available for all ATM/POS transactions.
MccCode	String{6}	Yes	Merchant Category Code
MerchantId	String{20}	Yes	Merchant Store Identification number
MerchantName	String{100}	Yes	Merchant Name & Location
TerminalId	String{15}	Yes	Terminal Id
MerchLocation	String{10}	Yes	Merchant Location
MerchCountryCode	String{3}	Yes	Merchant Country Code
PosEntryMode	String{25}	Optional	Authentication Type, i.e. CNP, Magstripe, PIN etc.
ResponseCode	String{4}	Yes	Success / Decline Response Code for the transaction
ResponseReason	String{8}	Yes	Success / Decline Response Reason for the transaction

4.13 Debit/Credit the Account - APCRDR

This API is used to charge the card holder (debit/credit) account for do any balance adjustment or to apply any additional fee apart from product definition.

Debit/Credit Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
ProxyNumber	String{16}	Yes	Either Card Number or Proxy Number can be accepted. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String{6}	Yes	Product code for the account to be charged
Type	String {3}	Yes	Supported Transactions are Balance Adjustment.(BAT) Charge the fee.(FEE)
Flag	String {2}	Yes	DR/CR - Indicates Debit/Credit the amount in the account.
Amount	String{12}	Yes	Amount to be Credit/Debit from Account.
Currency	String{3}	Yes	Three digit ISO currency code for the Amount to be debit/Credit. If the currency and Account currency varies then system will apply the exchange rate.
Remarks	String{250}	Yes	Specify the remarks for transaction which will be printed in the statement.

Debit/Credit Response

Based upon the type of transaction and flag the amount will be debit/Credit from the account.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String{16}	Yes	Echoed back from the request.
ProductCode	String{6}	Yes	Echoed back from the request.
BalanceAmount	String{12}	Yes	Balance on the account after the debit/Credit.

4.14 Fund Transfer - APFDTR

This API can be used to transfer fund from one card account to another card account of two different Card holders. Check performed if the Cards are in „Active“ status and „Source“ and „Destination“ accounts are linked to two different Card holders. Based on the input values such as Card numbers, target and destination accounts and the currency rate the fund transfer will be performed.

Fund Transfer Request

Transfer amount should in Destination Account currency. If it is cross currency fund transfer then exchange rate will calculated and then amount will be debited from Source account.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ExchangeRate	String{12}	Optional	Exchange need not to set for Same currency transfer. For Cross Currency specify the currency rate if not currency rate will be taken from the system which already set through ExchangeRate API.
RequestXml/SourceAccount			
ProxyNumber	String{16}	Yes	Either Source Card Number or Proxy Number where the amount to be debit. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String{6}	Yes	Short code of the product code.
AccountCcy	String{3 }	Yes	Currency of the source card account from where the fund transfers to be initiated.
RequestXml/DestAccount			
ProxyNumber	String{16}	Yes	Either Destination Card Number or Proxy Number where the amount to be transferred. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String{6}	Yes	Short code of the product code.
AccountCcy	String{3}	Yes	Currency of the destination account where the fund to be transferred
Amount	String{12}	Yes	Amount to be Credited.

Fund Transfer Response

The amount will get transferred immediately on successful processing. If transaction got failed, the reason code will send in the response along with description.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ExchangeRate	String{12}	Yes	Exchange rate applied for amount calculation.
ResponseXml/SourceAccount			
ProxyNumber	String{16}	Yes	Echoed back from the request.
ProductCode	String{6}	Yes	Echoed back from the request.
AccountCcy	String{3 }	Yes	Echoed back from the request.

Amount	String{12}	Yes	Amount debited from the source account(Excluding Fee charges)
FeeAmount	String{12}	Optional	Fee Amount if any charged on source account.
ResponseXml/DestAccount			
ProxyNumber	String{16}	Yes	Echoed back from the request.
ProductCode	String{6}	Yes	Echoed back from the request.
AccountCcy	String{3}	Yes	Currency of the destination account where the fund to be transferred
Amount	String{12}	Yes	Credited Amount.
FeeAmount	String {12}	Optional	Fee Amount if any charged on destination Account.

4.15 Exchange Rate - APFREX

This API can be used to update the exchange rates maintained in the Yalamanchili Platform. The buying and selling rates can be updated for a combination of Source and Destination currency.

- Check performed if the rates for the „Source“ and „Destination“ currencies are already available for the requested date.
- If the rates for the requested date are available then the values received in the request are updated, else the values are maintained in the System.
- Currency should be three digit ISO format. Ex for US dollars as USD.

Exchange Rate Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
RequestXml/ExRates/Detail - This node can be multiple when submitting more than one exchange rates detail			
BuyCurrency	String {3}	Yes	Currency code of the buying currency.
SellCurrency	String {3}	Yes	Currency code of the selling currency.
ExRate	String{12}	Yes	Exchange rate used for buying and selling.

Exchange Rate Response

Once the exchange is updated successful, transaction reference no is echoed back in the response along with response code.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.

4.16 Card Order - APCROR

This API can be used to place the Card Order for non personalized card type. The process for placing Card order as below

1. Card Order should place against the Branch in the System. So that Stock Inventory can be maintained.
2. In the request, if the branch id is not mentioned, then as default card order will be placed under the branch of API User.
3. As default, cards will be delivered to the Card Ordered branch. If requestor wants to deliver to different address, then specify the address in the Address filed.
4. Before placing the Order, the system will validate whether the product is available for the branch, Card design is available for the product. If those condition matched, then the order will be placed and invoice number will be generated and send in the response.
5. Use the invoice number for tracking the Order status.

Once the Card Order places then following process are involved with the entity Yalamanchili"s, Card Bureau and ServiceRequestor are

1. Yalamanchili will create the embossing file and send to the Card Bureau for Card Creation.
2. Only card has been created then Card Bureau will delivered the cards to the branch address or to the address mentioned in the card order request.
3. ServiceRequestor should receipt the card via "Card Receipt API-APSTRC" so that stock will get updated in the Narada.

Card Order Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
RequestXml/Order			
BranchId	String {8}	Optional	Branch ID of the Branch used for Card Order.
ProductCode	String {6}	Yes	Product code for the Card Order to be placed.
CardDesign	String{6}	Yes	Card Design for the order.
Quantity	String{10}	Yes	Number of cards to be ordered.
RequestXml/Order/Delivery - This filed can be optional, if not specified then card will be delivered to branch address of the order requestor. i.e branch of the api user.			
Name	String{50}	Yes	Name of the Branch
Address1	String{35}	Yes	Branch Address Line No1
Address2	String{35}	Yes	Branch Address Line No2
Address3	String{35}	Optional	Branch Address Line No3
Address4	String{35}	Optional	Branch Address Line No4
City	String {20}	Yes	Branch address City Name
State	String {20}	Yes	Branch address State Name
Country	String {20}	Yes	Branch address Country Name
PostalCode	String {15}	Yes	Branch address Postal Code

Card Order Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
InvoiceNo	String{20}	Yes	Invoice Number for the Card Order Placed. Use this Invoice number for order Tracking
OrderRefNo	String{10}	Yes	Reference for the Order. Use this reference number for Receipt the order /to report card damage

4.17 Card Receipt - APSTRC

Once the Card delivered to branch, then branch should acknowledge the delivery of the card by calling this API. So that in the Narada system cards will be marked as receipt and now it's ready for sale. Order reference number and pack reference is mandate to receipt the card. If there is any damage in the card then use the update stock API to report the damage and update the inventory too. Every card/Item is dispatched with pack reference no. Use this pack reference number in the request to confirm the receipt.

Card Receipt Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
RequestXml/Receipt			
OrderRefNo	String {10}	Yes	Order Reference Number.
Quantity	String {10}	Yes	No of Cards Received.
FromPackNo	String {12}	Yes	From pack Reference Number.
ToPackNo	String {12}	Yes	To Pack Reference Number.

Card Receipt Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the response.
OrderRefNo	String {12}	Yes	Echoed back from the response.

4.18 Update Stock Details - APSKUP

This API is used to update the stock if there is any damage in the card.

Update Stock Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
OrderRefNo	String {12}	Yes	Order Reference number for damage to be reported
Quantity	String {10}	Yes	Number of Cards damaged.
RequestXml/RefNumbers - Pack No field will be multiple depends no of Quantity get damaged.			
PackNo	String {12}	Yes	Pack Reference of the damaged Card.

Update Stock Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request
OrderRefNo	String {12}	Yes	Echoed back from the request
Quantity	String {10}	Yes	Echoed back from the request

4.19 Get Stock Details - APGTSD

This API is used to get the current stock status belong to branch requested. In the response cards ordered, card Activated, card with the branch (yet to activated).

Get Stock request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor.
BranchId	String {8}	Yes	Get the stock details of the branch requested.

Get Stock response

The following table list the fields are returned in the response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
BranchId	String {8}	Yes	Echoed back from the request.
ResponseXml/StockInfo - This node will be based on the product mapped to the branch.			
ProductCode	String {6}	Yes	Short code of the product
CardOrdered	String {10}	Yes	Numbers of ordered by the branch up to now for that product

CardActivated	String {10}	Yes	Number of activated till now.
CardWithBranch	String {10}	Yes	Number of cards yet to activate.
CardDamaged	String {10}	Yes	Number of cards will be damaged.

4.20 Status Request - APSTRQ

When the ServiceRequestor have not received the response from the Narada-API interface for initiated transaction, then this API will be helpful to know the status of transaction using original transaction reference Number.

Status Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Original Transaction reference Number

Status Response

Based on the response code returned in the response, ServiceRequestor will know whether transaction is received at the Narada-API Interface and processed. If the response code is “OO” then transaction is processed successful. If the transaction not received by Narada-API then system will return the response code as “TXNNRV” - Transaction not received. If the transaction received and processed unsuccessful then relevant response xml, code and description of the original will be returned in the response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.

4.21 Profile Creation - APRFCR

This API is used to create a user profile in the System. Upon user profile creation, the consumer/Card Holder can able to access his account via Mobile and Portals. Before calling Profile API, card should be in active.

Profile Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
UserId	String {8}	Yes	User Id for the profile access.

Password	String	Yes	Password should be minimum of 8 and maximum of 16 characters. Hash the password with SHA-256 and encode to BASE64 format then send to Narada-API.
EmailId	String { 50 }	Optional	Email ID of the Card Holder
MobileNumber	String { 15 }	Optional	Customer"s Mobile Number Format: (+) with Country Code.

Profile Response

If the requested UserID is available to create then system will creates the user and sends the successful response. The following table list the fields are returned in the response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.

4.22 Change Password - ACHPSS

This API is used to change the password. API should validate new password and retype new password after that call the API with old and new password.

Change password Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Unique Transaction Reference generated by ServiceRequestor.
OldPassword	String	Yes	Password should be minimum of 8 and maximum of 16 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.
NewPassword	String	Yes	Password should be minimum of 8 and maximum of 16 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.

Change password Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Echoed back from the request.

4.23 Access Rights - APACRT

This API is used to provide the list of functions which are allowed for User Profile. Note this API is applicable to retrieve only registered Card Holder Profile Information.

Access Rights Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
UserId	String {8}	Yes	User Id for the profile rights to be returned.

Access Rights Response

The following table list the fields are returned in the response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
UserName	String {100}	Yes	Name of the Card Holder
LastLogin	String {25}	Yes	Last Login time of the Customer with the system. Format DD-MON-YYYY HH:MI:SS AM/PM
ResponseXml/MenuList - This Node will have the list of Menu/Rights for the requested User			
TMenu	String {50 }	Yes	Name of the Menu
SMenu	String {50}	Yes	Sub Menu of the Main Menu

4.24 PIN Retrieval - APIPIR

This API is used to retrieve ATM pin from the System. The ATM PIN would have already got generated in our system during the card data generation and also stored in the Chip.

PIN Retrieval Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
ExpiryDate	String {4}	Yes	Expiry Date of the Card Front panel. In MMY format
CVV2	String {3}	Yes	CVV2 of the Card Back Panel.
DOB	String { 8 }	Yes	Customer" s Date of Birth. Format : DDMMYYYY

PIN Retrieval Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.
EPinBlock	String {16}	Yes	Encrypted PINBLOCK <u>Note1:</u> - It will always be a 4 digit numeric PIN. - The PINBLOCK will be in ISO FORMAT-0 - T-DES encryption would be applied on the PINBLOCK - We will use a static Zone Pin Key for encryption of the PINBLOCK. <u>Note2:-</u> The Key Management for this Zone Pin Key would have to be carried out for both Test and Production between the two parties (YALAMANCHILI and client) as per Standard Information Security Policies.

4.25 ATM PIN Verification - APIVER

This API is used to verify ATM pin from the System.

ATM PIN verification Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
EPinBlock	String {16}	Yes	Encrypted PINBLOCK <u>Note1:</u> - It will always be a 4 digit numeric PIN. - The PINBLOCK will be in ISO FORMAT-0 - T-DES encryption would be applied on the

			PINBLOCK We will use a static Zone Pin Key for encryption of the PINBLOCK. <u>Note2:-</u> The Key Management for this Zone Pin Key would have to be carried out for both Test and Production between the two parties (YALAMANCHILI and client) as per Standard Information Security Policies.
DOB	String { 8 }	Yes	Customer"s Date of Birth. Format : DDMMYYYY

ATM PIN verification Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.26 ATM PIN Change - APICHG

This API is used to change the ATM pin from the System.

ATM PIN change Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
EoldPinBlock	String {16}	Yes	Encrypted PINBLOCK <u>Note1:</u> - It will always be a 4 digit numeric PIN. - The PINBLOCK will be in ISO FORMAT-0 - T-DES encryption would be applied on the PINBLOCK - We will use a static Zone Pin Key for encryption of the PINBLOCK.

			<u>Note2:-</u> The Key Management for this Zone Pin Key would have to be carried out for both Test and Production between the two parties (YALAMANCHILI and client) as per Standard Information Security Policies.
EnewPinBlock	String {16}	Yes	Encrypted PINBLOCK <u>Note1:</u> - It will always be a 4 digit numeric PIN. - The PINBLOCK will be in ISO FORMAT-0 - T-DES encryption would be applied on the PINBLOCK - We will use a static Zone Pin Key for encryption of the PINBLOCK. <u>Note2:-</u> The Key Management for this Zone Pin Key would have to be carried out for both Test and Production between the two parties (YALAMANCHILI and client) as per Standard Information Security Policies.

ATM PIN change Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.27 Assign override settings - AOVRST

This API is used to campaign code mapped in the System. Campaign Code is a code used to identify a group of limits and fees. An example of Campaign Code would be special or differential fee conditions and limits to be applied for a VIP Personnel. The Campaign Code will be an over-ride to the product settings on limits and fees. The Campaign Code is assigned to one product mapped to the card number.

Pre-Requisite:

- Please be informed that the Campaign Code should already be pre-defined in the system using the Product Settings feature of the NARADA Prepaid Portal using the over-ride settings options.

Assign override settings Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
CampaignCode	String {10}	Yes	Campaign Code
ProductCode	String {6}	Yes	Short code of the product
Flag	String {1}	Yes	Campaign Code flag Enable/Disable [Y/N]

Assign override settings Response

All the Campaign Codes have an effective date(Start-Date and End-Date) and if the current date is within the Start-Date and End-date range only then the transaction is considered successful.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.

4.28 Alert Settings - set/unset - APALTS

The following API is used to configure the SMS or Text Message alert settings for a specific set of transactions to a given cardholder.

Alert settings Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
ProductCode	String {6}	Yes	Short code of the product
AlertType	String {6}	Yes	This is a string to represent the scenario or business transaction and can be any one of the following values 1. ATMCWD - This is for ATM Cash Withdrawal 2. PURCSE - This is for Purchases.

			3. LOWBAL - This is when the cardholders account goes below the configured value by the cardholder. 4. TOPACC - This is sent when the card account is topped up.
Frequency	String {3}	Yes	It can be either of the following values 1. DLY - Applicable only for LOWBAL 2. WLY - Applicable only for LOWBAL 3. MLY - Reserved for Future Use 4. YLY - Reserved for Future Use 5. TXN - Applicable for ATMCWD, PURCSE, TOPACC. Also this is a transaction triggered message if the transaction was successful.
AlertData	String{12}	Optional	This will contain the amount for a LOW BALANCE ALERT in the format 000000000200 if the LOW BALANCE value is set as 2 units of the cardholder currency
Channel	String{5}	Yes	Alert Type [SMS/EMAIL/BOTH]
AlertSet	String{1}	Yes	Alert set or unset [Y/N]

Alert settings Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.29 Get Card number - APIGCN

This API can be called to retrieve the card number of the card holder.

Get Card number Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.

Reftext1	String	Conditional	Future Use, This is based on client specific.
Reftext2	String	Optional	Future Use
Reftext3	String	Optional	Future Use

Get Card number Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.
CardNumber	String {16}	Yes	Card number will be present linked to ProxyNumber.
ExpiryDate	String {8}	Yes	Expiry Date of Customer"s Identification Proof Format :DDMMYYYY
NameOnCard	String {25}	Optional	Customer Preferred Name on the Card. This filed is applicable for Purchasing a personalized Card.

4.30 Get CSC Enquiry - ACSENG

This API can be called to retrieve the 4CSC. This functionality is only applicable for American Express Card Scheme - Virtual Cards.

Get CSC Equiry Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
ExpiryDate	String {4}	Yes	Expiry Date of Customer"s Identification Proof Format :YYMM

Get CSC enquiry Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.
CSC	String {4}	Yes	4CSC value will be set in the Response for the linked Card.

4.31 Card Order Status - APCOST

This API can be called to retrieve the status of card order.

Card Order Status Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
InvoiceNo	String {13}	Yes	Either Invoice Number or Order Reference Number can be given in this field.

Card Order Status Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
InvoiceNo	String {13}	Yes	Echoed back from the request.
InvoiceDate	String {8}	Yes	Invoice Date. Format :DDMMYYYY
OrderedBy	String {100}	Yes	Who is the initiate card order
ProductCode	String {6}	Yes	Short code of the product
PlasticName	String {100}	Yes	Plastic name of the card order
CardOrdered	String {10}	Yes	Numbers of ordered by the branch up to now for that product
OrderStatus	String {25}	Yes	Current Status of the card order

4.32 IVR PIN Set - APIIVR

This API is used to set the IVR pin. The IVR PIN would have already got generated in our system during the card data generation.

IVR PIN Set Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
IvrPin	String	Yes	IvrPin should be 6 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.

DOB	String { 8 }	Yes	Customer"s Date of Birth. Format : DDMMYYYY
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IVR PIN Set Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.33 IVR PIN Change - APIIVC

This API is used to change the IVR pin from the System.

IVR PIN change Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
oldIvrPin	String {16}	Yes	IvrPin should be 6 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.
NewIvrPin	String {16}	Yes	Pin should be 6 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.

IVR PIN change Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.34 IVR PIN Validation - APIPNV

This API is used to validate the IVR PIN from the System.

IVR PIN Validation Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
IvrPin	String {16}	Yes	IvrPin should be 6 characters. Hash the password with SHA-256 and encode to BASE64 format then send to NaradaPPH-API.
DOB	String { 8 }	Optional	Customer" s Date of Birth. Format : DDMMYYYY

IVR PIN Validation Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.35 Reset Web Password - APFORP

This API is used to reset web password. It is used to web portal only and not supported in API login credentials

Reset Web Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
Reftext1	String {100}	Optional	Future Use
Reftext2	String {100}	Optional	Future Use
Reftext3	String {100}	Optional	Future Use
NewPassword	String	Yes	Password should be minimum of 8 and maximum of 16 characters. Hash the password with SHA-256 and encode to BASE64 format then send to Narada-API.
ConfirmPassword	String	Yes	Password should be minimum of 8 and maximum of 16 characters. Hash the password with SHA-256 and encode to BASE64 format then send to Narada-API.

Reset Web Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.

4.36 ATM PIN Set - APINST

This API is used to set the ATM pin from the system.

ATM PIN Set Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
ExpiryDate	String {4}	Yes	Expiry Date of the Card Front panel. In MMY format
CVV2	String {3}	Optional	CVV2 of the Card Back Panel.
DOB	String {8}	Yes	Customer" s Date of Birth. Format : DDMMYYYY
EPinBlock	String {16}	Yes	Encrypted PINBLOCK <u>Note1:</u> - It will always be a 4 digit numeric PIN. - The PINBLOCK will be in ISO FORMAT-0 - T-DES encryption would be applied on the PINBLOCK - We will use a static Zone Pin Key for encryption of the PINBLOCK. <u>Note2:-</u> The Key Management for this Zone Pin Key would have to be carried out for both Test and Production between the two parties (YALAMANCHILI and client) as per Standard Information Security Policies.

ATM PIN Set Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
ProxyNumber	String {16}	Yes	Echoed back from the request.

4.37 New OTP Request - AFPOTP

This API is used to new OTP generation.

New OTP Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Unique Transaction Reference generated by ServiceRequestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
ExpiryDate	String {4}	Optional	Expiry Date of the Card Front panel. In MMY format
Reftext1	String {25}	Optional	Future Use1, It is used to ServiceName.
Reftext2	String {25}	Optional	Future Use2
Reftext3	String {25}	Optional	Future Use3

New OTP Response

If the requested new OTP is sends to SMS. The following table list the fields are returned in the API response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Echoed back from the request.

4.38 OTP Validation - AVPOTP

This API is used to OTP validation.

OTP validation Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Unique Transaction Reference generated by ServiceRequestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
OTP	String {6}	Yes	OTP Value

OTP validation Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {12}	Yes	Echoed back from the request.

4.39 Retrieve Web Profile - APUSPF

This API is used to retrieve web profile like user id, mobile no, email. User Id is used to web portal only and not supported API login credentials

Retrieve Web Profile Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
ProxyNumber	String {16}	Yes	Either Source Card Number or Proxy Number. Card Number will be 16 digits and Proxy Number will be 12 digits.
Reftext1	String {100}	Optional	Future Use
Reftext2	String {100}	Optional	Future Use
Reftext3	String {100}	Optional	Future Use

Retrieve Web Profile Response

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
UserId	String {8}	Yes	User Id for the profile access.
EmailId	String { 50 }	Optional	Email ID of the Card Holder
MobileNumber	String { 15 }	Optional	Customer"s Mobile Number Format: (+) with Country Code.

4.40 Corporate Registration - ACRREG

This API is used to register a corporation registration in the YMC prepaid system.

Corporate Registration Request

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by Service Requestor.
CorpNameOne	String {100}	Yes	Corporate Name 1
CorpNameTwo	String {100}	Optional	Corporate Name 2

CorpNameThree	String {100}	Optional	Corporate Name 3
StreetName	String {100}	Yes	Street Name & Number
City	String {50}	Yes	City
Country	String {30}	Yes	Country
PostalCode	String {8}	Yes	Postal Code
Corporateld	String {10}	Optional	If the data is not populated, YMC will create an ID and assigns to corporate. YMC created ID Ex: J298N2083
CrmlId	String {20}	Optional	CRM Id
CorporateLoginId	String {10}	Yes	CorporateLoginId. It will be used to login into corporate portal
Enterpriseld	String {50}	Optional	Enterprise Id. If data is populated, just ensures that the corporate id and enterprise id are not same.
TaxNumber	String {15}	Optional	Tax Number
PrintName	String {20}	Optional	Print name
Optional1	String {100}	Optional	Optional future use
Optional2	String {100}	Optional	Optional future use
Optional3	String {100}	Optional	Optional future use
RequestXml/CorporateContact			
Gender	String{1}	Yes	Gender
Title	String {25}	Optional	Title. It is value should be any one of the following: Dr, Mr, Ms, Univ, Prof.
FirstName	String{30}	Optional	First name
LastName	String{30}	Yes	Last name
MailAddress	String{100}	Optional	Mailing Address
OfficePhone	String {20}	Optional	Office Telephone, It is accept numeric & '-'(delimiter) Only Format: Country Code-Area Code-Phone No
ResidentPhone	String {20}	Optional	Residential Phone, It is accept numeric & '-'(delimiter) Only Format: Country Code-Area Code-Phone No
Mobile	String {20}	Optional	Mobile, Format : countryCode+mobilenos, It is accept numeric & '-' (delimiter) Only Format: Country Code-Area Code-Phone No
Fax	String {20}	Optional	Fax, Format : countryCode+phone no, It is accept numeric & '-'(delimiter) Only Format: Country Code-Area Code-Phone No
Suburb	String {30}	Optional	Suburb
City	String {50}	Optional	City
State	String {30}	Optional	State
Country	String {30}	Optional	Country

EmailId	String {100}	Yes	Email Id. It is accepting alpha-numeric + special Char: ('-' '.' '_' '@')
PostalCode	String {8}	Optional	Postal code. It is accept alpha-numeric.
RequestXml/Corporate/Branch - This Node will be Mandatory.			
CorpBrNmOne	String {25}	Yes	Corporate Branch Name
CorpBrNmTwo	String {25}	Optional	Corporate Branch Name
CorpBrNmThree	String {25}	Optional	Corporate Branch Name
BranchCode	String {20}	Optional	Branch Code, if not give the BranchCode, YALAMANCHILI system will random branch code generate automatically.
BranchLoginId	String {10}	Yes	Branch Login
RequestXml/Corporate/Branch/ContactDetails			
Gender	String{1}	Yes	Gender, Value should be M F
Title	String {25}	Optional	Title. It is value should be any one of the following: Dr, Mr, Ms, Univ, Prof.
FirstName	String{30}	Optional	First name
LastName	String{30}	Yes	Last name
MailAddress	String{100}	Optional	Mailing Address
OfficePhone	String {20}	Optional	Office Telephone, It is accept numeric & '-' (delimiter) Only Format: Country Code-Area Code-Phone No
ResidentPhone	String {20}	Optional	Residential Phone, It is accept numeric & '-' (delimiter) Only Format: Country Code-Area Code-Phone No
Mobile	String {20}	Optional	Mobile, Format : countryCode+mobilenos, It is accept numeric & '-' (delimiter) Only Format: Country Code-Area Code-Phone No
Fax	String {20}	Optional	Fax, Format : countryCode+phone no, It is accept numeric & '-' (delimiter) Only Format: Country Code-Area Code-Phone No
Suburb	String {30}	Optional	Suburb
City	String {50}	Optional	City
State	String {30}	Optional	State
Country	String {30}	Optional	Country
EmailId	String {100}	Yes	Email Id. It is accept alpha-numeric + special Char: ('-' '.' '_' '@')
PostalCode	String {8}	Optional	Postal code. It is accept alpha-numeric.
RequestXml/LoadDetails			
ProductAssigned	String {100}	Yes	It will be a pipe separated - For the assignment or products
InvoiceLevel	String {3}	Optional	It is accept ALPHA characters For Ex : ENT,COR,BRN

PrefundingPrdCode	String {6}	Optional	This should be the prefunding product code. Based on this the prefunding product and account will be created for the corporate. If this field is empty, prefunding product and account number will not be created for the corporate
OverrideCode	String {6}	Optional	A corporate can be assigned multiple products using pipe separator. For Ex: MEAGR TOPUP (So these two products will be assigned) Each product can be assigned multiple override codes for a corporate. While assigning a override code to a card, there will be a validation to ensure the given code is one of the allowed code for that corporate or not Example: Assume, two override codes for each of the product showed in the above example. Then following should be a format. OVER01~OVER02 OVER03~OVER04 OVER1&OVER2 will be assigned to 1st product MEALGR. OVER3 & OVER4 will be assigned to second occurrence product TOPUP. ~ as product separator ~ as override code separator within the product.

Corporate Registration Response

Based on the above interface we need to create a corporate in the Yalamanchili Prepaid Application. If for the same corporate the webservice is called again we need to replace the current information with new corporate information and sends the successful response. The following table list the fields are returned in the response.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the request.
Corporateld	String {10}	Optional	Corporate Id
Enterpriseld	String {50}	Optional	Enterprise Id
BranchLoginId	String {10}	Optional	Corporate Branch Login Id

4.41 Add Card - APNADC

This API is used to purchase a supplementary card with the System.

Add card API is used to add the card for the existing account. Here the account can be access by the multiple cards. If the fee configured of the add card functionality, then the fee transaction take place

for the existing card. In the Non - personalized card request, new card number product should match with the product of the existing card number. BIN range between the two cards is not considerable.

Add Card Request

While initiating the Add Card transaction, then following parameters has to be sent to the Prepaid API System.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
RequestXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character length.
ProxyNumber	String {16}	Yes	Mandatory for Instant Card Registration.
NewProxyNumber	String {16}	Optional	Purchase new Proxy Number
CustomerId	String {100}	Yes	Customer Identifier generated by ServiceRequestor and used to map the customer of ServiceRequestor proprietary System with Yalamanchili Prepaid System.
NameOnCard	String {25}	Optional	Customer Preferred Name on the Card. This filed is applicable for Purchasing a personalized Card.
AddCardFlag	String {1}	Yes	S - Supplementary Card A - Add On Card Supplementary Card:- Newcard given in the request will map with existing account(Balance). New card will map with the account corresponding to the given ProductCode in this request. Add On Card :- New card given in the request will map with the new account (Balance)
CardDesign	String {6}	Optional	This field can be used to mention the Plastic Design required for the card. This is used for the personalized card issuance. If the Value not Present then Default Design will be in use.
CardActivationFlag	String {1}	Optional	Set to "Y" if cards need to be activate immediately and "N" to activate later. This field is applicable only for the Instant Card and Virtual Card. If the value is "N" then the activation code is generated and available in the New Card Response. This activation code is used to activate the card through the Card Holder Portal /IVR
RefText1	String {32}	Optional	For Future Use
RefText2	String {32}	Optional	For Future Use
RefText3	String {32}	Optional	For Future Use

RefText4	String {32}	Optional	For Future Use
RequestXml/LoadDetails			
ProductCode	String {6}	Yes	Short Code of product to map with the Card.
LoadAmount	String {12}	Yes	Amount to be loaded into the Card account of the card holder
BillingAmount	String {12}	Yes	This field contains the amount in the card account currency
LoadCurrency	String {3}	Yes	Currency used by the Card holder to fund the card account. For example, if Card holder used USD to load a GBP Card account, then Loading currency will be USD
BillingCurrency	String {3}	Yes	This is the currency of the card account

Add Card Response

The following table lists the fields returned in the response from the Narada-PPH API along with ResponseCode and ResponseText. Refer the Appendix for the Code and Information.

ELEMENT	DATA TYPE	REQUIRED	DESCRIPTION
ResponseXml - Root Node of the of XML			
TxnRefNo	String {32}	Yes	Echoed back from the Request
ProxyNumber	String {16}	Yes	Echoed back from the Request
NewProxyNumber	String {16}	Optional	For Personalized card it will be generated and set to the field. If it is Instant card then echoed back the Proxy Number.
ActivationCode	String {6}	Optional	6 digits Number will be set. As default Activation Code will be generated for Personalized Card. For Other Cards it will be based on Card Activation Flag.
CVV	String{3}	Optional	CVV2 value will be set in the Response only for the Virtual Card.
CardExpiry	String{8}	Optional	Expiry Date Value will be set to the response only for the Virtual Card.

5 IMPORTANT NOTIFICATIONS

1. All API request element TxnRefNo - Unique Transaction Reference generated by ServiceRequestor (12 character length) / external vendor (MASTER/VISA) 32 character lengths.
2. Request xml should be maintaining the sequence order as per API specification request element column.
3. ServiceRequestor will send API request should be HTTP - POST method only (YMC support only HTTP - POST).
4. Encryption/Decryption is mandatory for all API transaction. After implementation of encryption, we can start the UAT.

6 SECURITY IMPLEMENTATION

6.1 Certificates Management

1. Service Requestor should have a Key pair generated securely and share its certificate to Yalamanchili. The certificate should be shared in the Base-64 encoded binary X 509 (.CER) format.
2. Yalamanchili will designate a certificate custodian for receiving the certificate from the client. The Service Requestor should send the certificate only to the designated certificate custodian at Yalamanchili.
3. The Service certificate expiry period must not exceed 2 years.
4. The certificate must be signed by a private or public CA. The Root and Intermediate certificate (if applicable) should be shared in the Base-64 encoded binary X.509 (.cer) format.
5. Yalamanchili will share its certificate, Intermediate certificate and Root certificate in the Base-64 encoded binary X.509 (.cer) format.
6. Client needs to designate a certificate custodian and Yalamanchili will send its certificates only to the certificate custodian at client.

6.2 Keys & Certificates Usage

S. No	Owner	Stored At/Generated By	Key/Certificate	Usage
1	Client	Client	Private Key	Sign the API Request sent to Yalamanchili Decrypt the Session Key used to encrypt the API Response message
2	Client	Yalamanchili	Certificate	Verify the API Request received by Yalamanchili Encrypt the Session keys used to encrypt the API Response

				message.
3	Yalamanchili	Yalamanchili	Private Key	Sign the API Response sent to Client Decrypt the Session Key used to encrypt the API Request message
4	Yalamanchili	Client	Certificate	Verify the API Response received by Client Encrypt the Session keys used to encrypt the API Request message.
5	Yalamanchili	Yalamanchili	Session Key	To Encrypt the Response Messages sent to client
5	Client	Client	Session Key	To Encrypt the Request Messages sent to Yalamanchili

6.3 Encryption Algorithms

S. No	Type	Algorithm	Usage (Java)
1	Signature	SHA512withRSA	Signature.getInstance("SHA512withRSA", "BC");
2	Message Digest	SHA-256	MessageDigest.getInstance("SHA-512", "BC");
3	RSA Cipher	RSA/ECB/OAEPWITHSHA-256ANDMGF1PADDING	Cipher.getInstance("RSA/ECB/OAEPWITHSHA-256ANDMGF1PADDING", "BC")
4	AES Cipher	AES/GCM/NoPadding	Cipher.getInstance("AES/GCM/NoPadding", "BC")

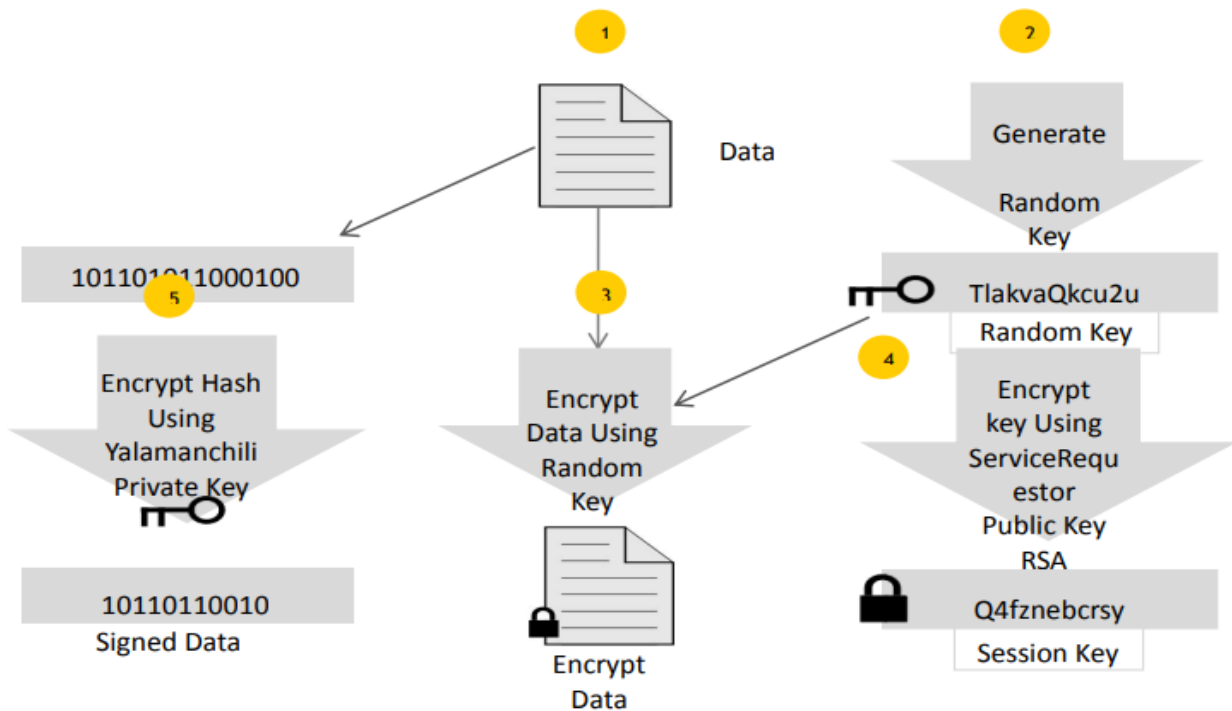
6.4 Key Strength

S. No	Key Type	Key	Usage
1	RSA Keys	2048 Bits	Signature and Session Key Encryption
2	AES Session Keys	256 Bits	Message Encryption

6.5 Other Security Considerations

1. Service Requestor will send API request should be HTTPS - POST method only (Yalamanchili support only HTTPS - POST).
2. Payload Encryption, integrity check and non-repudiation check are mandatory for API transactions.
3. Public IP validation is mandatory if the API is accessed via internet.
4. Yalamanchili advise the client to access APIs via IPsec VPNs or MPLS for enhanced security.
5. The java code snippet provided in this document is a general guideline and not a final version. It may differ based on the Operating System, JDK version, PKI provider used etc

6.6 Signing and Encrypting the Request



Step 1: Construct XML String as per the message specification. For example, we need to consider below message for Query Balance- APACBL.

```
<RequestXml><ProxyNumber>4444444444444444</ProxyNumber><ProductCode>GFTCRD</ProductCode></RequestXml>
```

Note: The XML string should be in single line without indentation and without spaces in between the xml tags

Step 2: Generate symmetric key with key size of 256 bits (32 bytes) using AES Algorithm. The random key should be generated for every transaction.

```
KeyGenerator kgen = KeyGenerator.getInstance("AES");
kgen.init(256);
SecretKey key = kgen.generateKey();
return key.getEncoded();
```

Step 3: Encrypt the constructed XML string using session key and convert the encrypted data as Base64 encoded string. Retrieve the <msgRefNo> field data having length of 12 bytes from the request header and append four zeros (0000) to make it to a length of 16 bytes and use as Initialization Vector (IV). This is included in the <request> field in the requestBody of the SOAP XML message.

msgRefNo - This is the msgRefNo used in the request header.

sessionKey - Session key generated in the Step 2

strIv - This is the initialization vector derived from msgRefNo by padding four zeros.

xmlRequestData - This is the XML string to be encrypted using the session key.

```
StringBuilder strlv = new StringBuilder(msgRefno); // pass same in request header
strlv.append("0000");
SecretKeySpec symmKeySpec = new SecretKeySpec(sessionKey, "AES");
Cipher symmCipher = Cipher.getInstance("AES/GCM/NoPadding");
symmCipher.init(Cipher.ENCRYPT_MODE, symmKeySpec,
new GCMParameterSpec(16 * Byte.SIZE, strlv.toString().getBytes()));
return Base64.encodeBase64String(symmCipher.doFinal(xmlRequestData));
```

Step 4: Encrypt the generated session key using Yalamanchili's public key and convert the output to Base64 encoded string. This is included in the <sessionKey> field in the requestHeader of the SOAP XML message.

pubKeyFileName - This is the Yalamanchili public key file name

sessionKey - This is the session key created in the step 2

```
PublicKey pubKey = getPublicKey(pubKeyFileName, "RSA");
Cipher encryptCipher = Cipher.getInstance("RSA/ECB/OAEPWITHSHA-256ANDMGF1PADDING");
encryptCipher.init(Cipher.ENCRYPT_MODE, pubKey);
encryptionByte = encryptCipher.doFinal(sessionKey);
return base64Encode(encryptionByte);
```

Step 5: Sign the constructed XML string with Service Requestor's (Client) private key and convert the output to Base64 encoded string. This is included in the <token> field in the requestHeader of the SOAP XML message.

pvtKeyFileName - Client private key file name

dataToSign - This is the XML string to sign.

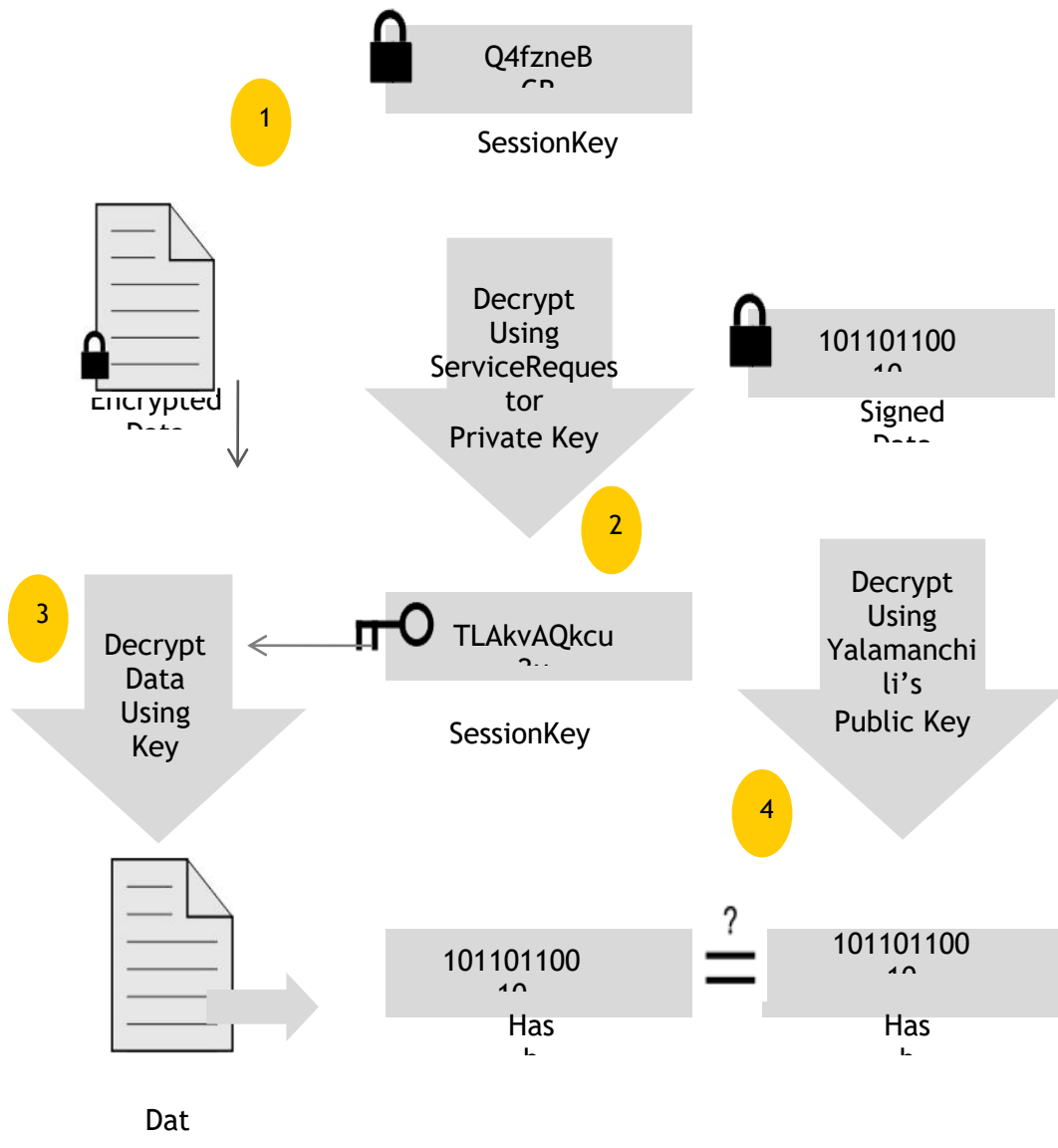
```
PrivateKey privateKey = getPrivateKey(pvtKeyFileName); Signature signature =
Signature.getInstance("SHA512withRSA");
signature.initSign(privateKey);
signature.update(dataToSign);
return Base64.encodeBase64String(signature.sign());
```

Step6: Construct the SOAP message and place the encrypted XML string, signed message and encrypted session key in the respective fields of the SOAP message as below

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://services.ycs.com"
xmlns:xsd="http://model.ycs.com/xsd">
  <soapenv:Header/>
  <soapenv:Body>
    <ser:serviceCall>
      <ser:naradaRequest>
        <xsd:requestHeader>
          <xsd:apiUserId>String</xsd:apiUserId>
          <xsd:apiPasswd>String</xsd:apiPasswd>
          <xsd:apiKey>String</xsd:apiKey>
          <xsd:channelType>String</xsd:channelType>
          <xsd:clientId>String</xsd:clientId>
          <xsd:msgRefNo>String</xsd:msgRefNo>
          <xsd:msgTimeStamp>String</xsd:msgTimeStamp>
          <xsd:sessionKey> {Place the Encrypted Session key} </xsd:sessionKey>
          <xsd:token> {Place the Signed Data} </xsd:token>
        </xsd:requestHeader>
        <xsd:requestBody>
          <xsd:request> {Place the Encrypted XML string} </xsd:request>
```

```
</xsd:requestBody>
</ser: naradaRequest>
</ser:serviceCall>
</soapenv:Body>
</soapenv:Envelope>
```

Verifying and Decrypt the Response



Step 1: Parse the Soap Message and read the Encrypted Data, Signed Data and Encrypted Session key from fields of sessionKey, token and response respectively.

```
soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ser="http://services.ycs.com"
xmlns:xsd="http://model.ycs.com/xsd">
  <soapenv:Header/>
  <soapenv:Body>
    <ser:serviceCallResponse>
      <ser:naradaResponse>
        <xsd:responseHeader>
          <xsd:apiKey>String</xsd:apiKey>
          <xsd:channelType>String</xsd:channelType>
          <xsd:clientId>String</xsd:clientId>
          <xsd:msgRefNo>String</xsd:msgRefNo>
          <xsd:msgTimeStamp>String</xsd:msgTimeStamp>
          <xsd:sessionKey> {Encrypted Session key} </xsd:sessionKey>
          <xsd:token> {Signed Data } </xsd:token>
        </xsd:responseHeader>
        <xsd:responseBody>
          <xsd:responseCode>String</xsd:responseCode>
          <xsd:responseText>String</xsd:responseText>
          <xsd:response> {Encrypted XML string} </xsd:response>
        </xsd:responseBody>
      </ser:naradaResponse>
    </ser:serviceCallResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Step 2: Decode the SessionKey data from Base64 format and then decrypt the output with ServiceRequestor's Private Key .Supported Cipher Algorithm Name/ MODE/ Padding for decrypting the symmetric key is RSA/ECB/PKCS1Padding.

Step 3: Decode the Encrypted data from Base64 format and then decrypt the output with Symmetric Key to derive clear xml response. Supported Cipher Algorithm Name/ MODE/ Padding for decrypting the symmetric key is AES/CBC/PKCS5Padding.

Step 4: Decode the Signed data from Base64 format and then decrypt the output with Yalamanchili's Public key to obtain Hash Data and since hashing is a one-way process so re-evaluate the hash using the algorithm SHA1withRSA. Then both have been compared and if there is a match, the signature has verified and ServiceRequestor's accept the message as coming unaltered from the Yalamanchili.

7 APPENDIX

7.1 Transaction Summary - APIkey

S.No	API Name	API Key
1	Purchase a Card/Register the customer	APNCRD
2	Update Customer Details	APUPCR
3	Get Card Details	APCHDT
4	Block / Lock / Unlock the Card	ACDSTS
5	Card Activation	APCDAC
6	Card Replacement	APCDRP
7	Account Addition	APACAD
8	Load Card / Funding an Account	APLCRD
9	Query Balance	APACBL
10	Amount Refund	APCRND
11	Account closure	APACCL
12	Get X Transactions	APXTXN
13	Debit/Credit the Account	APCRDR
14	Fund Transfer	APFDTR
15	Exchange Rate	APFREX
16	Card Order	APCROR
17	Card Receipt	APSTRC
18	Update Stock Details	APSKUP
19	Get Stock Details	APGTSD
20	Status Request	APSTRQ
21	Change Password	ACHPSS
22	Profile Creation	APRFCR
23	Access Rights	APACRT
24	PIN Retrieval	APIPIR
25	ATM PIN Verification	APIVER
26	ATM PIN Change	APICHG
27	Assign Override settings	AOVRST
28	Alert settings - set/unset	APALTS
29	Get card number	APIGCN
30	Get CSC Enquiry	ACSENQ
31	Card Order Status	APCOST
32	IVR PIN Set	APIIVR
33	IVR PIN Change	APIIVC
34	IVR PIN Validation	APIPNV

35	Reset Web Password	APFORP
36	ATM PIN Set	APINST
37	New OTP Request	AFPOTP
38	OTP Validation	AVPOTP
39	Retrieve Web Profile	APUSPF
40	Corporate Registration	ACRREG
41	Add Card	APNADC

7.2 Card Status

Card Status	Description
I	Active
CN	Card Returned
M	Card Issued
C	Closed
L	Locked
B	Blocked
HL	Hot List Lock
HS	Hot List Stolen
HD	Hot List Damaged
D	Delivered
CL	Cancelled
Q/K	Pin Regeneration requested
E	Expired
FL	Fraud lock

7.3 List of Error Codes

Response Code	Response Text
00	Transaction Success
03	Unable to process the request
09	Transaction amount is greater than card account balance
10	Invalid Card No
11	Error While Extracting Message Format
17	No records found
20	Unable to process the request
45	Card is in Closed status/TXN AMOUNT should not be in negative
47	Card is Not Active
103	Invalid CVV/ExpireDate/DOB
104	Invalid EPinBlock
147	Card Balance is Negative
177	Unable to process the request

213	Not enough Records for Refund Request
214	Not enough Records for Top UP Request
217	Invalid Customer Id
412	This operations is not allowed on Suspended card
1777	Unable to process the request
TXNNRV	Transaction not received
ACCCLS	All Accounts Closed For the Card
ACCEXT	Account Already Existed With the Currency.
ACTCRD	Active Cards Cannot Be Replaced
AGNRST	This is not valid customer for this agency
ALRCLD	Card is Already Closed or Cancelled
AMTEXE	Transfer Amounts Exceeded more than Block Amount.
BALNT0	Balance must be Equal to Zero
CACBLK	Card account is in Blocked status
CACINA	Card account is not in active status
CACLCK	Card account is in Locked status
CAPCHA	Checking Limit for Card
CARDBL	Card is Blocked
CARDLC	Card is in Locked State
CARDLC	Card is Closed
CAREXD	Card is Expired
CBELIM	Card Balance exceed the Limit
CBNTCG	Card Balance Limit Configuration Is Not Done
CCDNOT	Campaign Code Does Not Exist
CIDEXT	Customer Already Exist
CIDEXT	Customer Id Already Exists
CLACCT	This Operation cannot be performed
CRDBLP	Permanent Blocked, Card Status Cannot Be Changed
CRDBLO	Card is Blocked
CRDCLD	Card is Closed
CRDEXP	Card Expired
CRDINA	Invalid Card Number/Card is Inactive
CRDLKD	Card is Locked
CRDLOK	Card is in Blocked status
CRDNCP	Card is not belonging to the Corporate
CRPDWG	Current Password is Wrong
DENY	Error While Extracting Message Format
DMLERR	Error in DML Operations
ENTBLK	Entity is blocked
ENTCDE	Error In Entity Credit Days Check

EXEAMT	Refund Amount is Greater than Balance
FINERR	Error in financial Block
INACRD	Inactive Card
INCRCN	Card No Does not belong to this Corporate Id
INNCRD	Card is In Active
INTERR	Unable to process the request
INVACC	Access Denied - Rights not Assigned
INVADR	Invalid Address Format
INVCID	Invalid Customer ID
INVCRD	Invalid Card Number/Proxy Number
INVCST	Invalid Customer
INVOPR	This operations is not allowed on Cancelled cards
INVPLA	Invalid Card Design
INVREF	Invalid Reference Number for the Card and Amount Blocked.
INVTRN	Invalid Transaction Code
INVTXN	Transaction code is not configured
INVUSR	Invalid User
IVRNEQ	Both IVRPIN are not equal
LMAMDL	Per day limit exceeded.
LMAMTE	Per transaction limit exceeded.
LMANEX	Annual limit Exceeded
LMDLEX	Daily limit Exceeded
LMHFEX	Half yearly limit Exceeded
LMLTEX	Lifetime limit Exceeded
LMMTEX	Monthly limit Exceeded
LMQUEX	Quarterly limit Exceeded
LMTNOT	None of the Limit Codes are configured for that Product.
LMWKEX	Weekly limit Exceeded
LSSBAL	Available Card Balance is Less than the Amount
MATCRD	Card Status is Same. So, Cannot Be Changed
NEGBAL	Card is in Negative Balance
NOCHRY	No Records found for the Inputs Card No and Currency
NOCLET	No Client
NOCUST	Customer is not Registered.
NORCRA	No records available for the requested card number
NORCRB	No Records Found
NORCRC	No Records Found
NOTACT	Card is In Active
NOTAUT	Not Authorized
NOTINV	Card is not part of inventory

NOTRVP	New Wallet for the Currency Not available - Cannot Assign Product
ORDEXT	Replacement Card Order already exists on this card
PIDNXT	No Products Exists
PSDCNT	Password cannot be changed within seven days
PSTHIS	Password should not be in last 13 passwords.
REFDAT	Data Exists within the 10 Days in MAS_CHRGTBH_HISTORY
RPLFLD	Both The cards Must be Blocked
SPLERR	Error in splitting Input Data
SUPLER	This Operation is Not Allowed In Presence of Supplementary Card one
SYSERR	System Error
TASRMA	Transaction Amount should be more than or equal to Min. Amount
TIDEXT	Transaction Reference Already Existed
TXNALT	Transaction Reference Number is already exists
TXNBLK	Channel is Restricted
USMPSD	New Password should not be the part of user id
INVAPI	Invalid apiKey
INVALIDPASSWORD	Invalid login credential
INVDSG	Invalid Digital Signature
INVBRN	Invalid Branch User
INVUSR	Invalid login credential
USRBLK	User is blocked
USRSUS	User is suspended
USRCLO	User is closed
EXPPWD	Password is expired
NOCVV2	No CVV found for this card
XTRACT	Invalid Branch User
NOPRDC	No Product Count
NOZERO	Card Balance is greater than Zero
NGTBAL	Card Balance is Negative
INVEXR	Invalid Exchange Rate
INVPRD	Invalid Wallet
NOEXRT	Exchange Rate is not Defined
INMAAD	Invalid mailing address
INRSAD	Invalid Residence address
ERRTOP	Error During TopUp
INPDID	Invalid product mapping
CRDACT	Card is Already Activated
REQENC	Required session key and encryption request
REQTOK	Required token

INV DAT	Invalid input for From Date/To Date
USREXT	User ID Is Already Existed
INBRID	Invalid Branch ID
CRDCLO	Card is Closed
ACCCLD	Account is Closed
IACTNO	Invalid Activation Number
NTBLCD	Only Blocked Card can be Replaced
NTGAMT	Amount should not be in Negative
CRDHTL	Card is Hot Listed
NOTSAM	Source and Destination Card Number should not be same
SRCACC	Source Card Account should be Active
CBLZRO	Card Balance should not be ZERO
ZETRAM	Transfer Amount should not be Negative or Zero
DESACC	Destination Card Account should be Active
TXNINA	Channel has been restricted
INCDUS	Invalid Card User
INVCTR	User should be Customer
DCCNER	Invalid identification type request
AGELMT	Customer Age should be greater than 18
INVCCY	Bill Ccy should be equal to Product Ccy
NOTFRT	Already user profile created
INACFL	Activation Flag should not be empty for this request
INV DAT	Invalid Date
INPDID	Invalid Product Mapping
VITPRD	Virtual Product should not be allow here
INVQTY	Invalid Card Quantity
INV FNO	Invalid FromPackNo
INVTNO	Invalid ToPackNo
INVETY	Invalid Entries
INVEXR	Invalid Exchange Rate
BUYNOT	Buy Currency Code Not Exists
SELNOT	Sell Currency Code Not Exists
INVCMP	Invalid Campaign Code
INCPDT	Campaign Code Effective Date is Greater than Current_Date
INVFRQ	Invalid Frequency
IVALDA	Alert Data is Must For LOW BALANCE ALERT
ACTEXT	Account Already Exist for the Card
DATERR	Invalid Data
CRDTMB	Card is Temporarily Blocked
INVFCY	Invalid Source account currency

INVFPD	Invalid Source card account
INVTPD	Invalid Destination card account
INVTCY	Invalid Destination account currency
INVEXP	Invalid exchange rate for card account
DATER1	From_date Should not be Greater than To_date
DATER2	From_date Should not be Greater than Current_date
DATER3	From_date Should not be 6 Months older than Current_date
DATER4	To_date Should not be Greater than Current_date
DATER5	Date Difference between from_date and to_date should not be greater than 90
CRATMB	Card is Already Temporarily Blocked
CRABLP	Card is Already Permanently Blocked

7.4 List of Transaction Codes

These are generalized transaction types for common API specification. It is changed through client based.

Transaction Code	Transaction Name+
000000	Purchase - Non - WEB [Domestic Same Currency]
000001	Purchase Fee- Non - WEB
000002	Purchase- Non - WEB [Domestic Cross Currency]
000003	Purchase- Non - WEB [International Same Currency]
000004	Purchase- Non - WEB [International Cross Currency]
000005	Purchase- Non - WEB[Domestic Cross Currency based on Billing ccy authorization]
000006	Purchase- Non - WEB[International Cross currency based on Billing ccy authorization]
000020	Decline Purchase- Non - WEB
000021	Decline Purchase Fee- Non - WEB
010000	ATM Cash Withdrawal- Non - WEB [Domestic Same Currency]
010002	ATM Cash Withdrawal- Non - WEB[Domestic Cross Currency]
010003	ATM Cash Withdrawal- Non - WEB[International Same Currency]
010004	ATM Cash Withdrawal- Non - WEB[International Cross Currency]
010005	ATM Cash Withdrawal- Non - WEB[Domestic Cross Currency based on Billing ccy authorization]
010006	ATM Cash Withdrawal- Non - WEB[International Cross currency based on Billing ccy authorization]
010010	ATM Cash Withdrawal- Non - WEB [Not In Use]
010020	Decline CWD- Non - WEB

010021	Decline CWD Fee- Non - WEB
011101	Declined ATM Transaction- Non - WEB [Not Used in PPH]
011102	Declined ATM Transaction- Non - WEB[Not Used in PPH]
013000	Cash Advance- Non - WEB
020000	Adjustment Debit- Non - WEB
020001	Adjustment Debit- Non - WEB
070000	ATM Cash Withdrawal- Non - WEB [SMS Domestic Same Currency]
070002	ATM Cash Withdrawal- Non - WEB[SMS Domestic Cross Currency]
070003	ATM Cash Withdrawal- Non - WEB[SMS International Same Currency]
070004	ATM Cash Withdrawal- Non - WEB[SMS International Cross Currency]
070005	ATM Cash Withdrawal- Non - WEB[SMS Domestic Cross Currency based on Billing ccy authorization]
070006	ATM Cash Withdrawal- Non - WEB[SMS International Cross currency based on Billing ccy authorization]
070011	Branch Fund Transfer - WEB
080000	Purchase- Non - WEB[SMS Domestic Same Currency]
080002	Purchase- Non - WEB[SMS Domestic Cross Currency]
080003	Purchase- Non - WEB[SMS International Same Currency]
080004	Purchase- Non - WEB[SMS International Cross Currency]
080005	Purchase- Non - WEB[SMS Domestic Cross Currency based on Billing ccy authorization]
080006	Purchase- Non - WEB[SMS International Cross currency based on Billing ccy authorization]
090000	Balance Inquiry- Non - WEB [SMS Balance Inquiry]
100000	Pin Verification- Non - WEB
100001	Customer Registration- WEB
100003	Customer Registration FeeWEB
100004	Subscription Fee
100005	Customer Registration Fee
100006	ATM Cash Withdrawal Fee-- Non - WEB
100007	ATM Declined Fee-- Non - WEB
100008	ATM Declined Fee- Non - WEB
100009	Agency Registration- WEB
100010	Council Registration - WEB
100011	Lost Card - WEB
100012	Lost Card Fee - WEB
100013	Card Suspend/Unsuspend - WEB
100014	Card Suspend/Unsuspend Fee- WEB
100015	Employee Registration- WEB

100016	Customer Registration-WEB
100017	IPIN Request-WEB
100018	IPIN Request Fee-WEB
100020	Personalised Cus Registration-WEB
100021	Product Settings-WEB
100022	Card Statement-WEB
100023	Card Statement Fee-WEB
100024	Delivery Fee-WEB
100025	Branch Card Issuance-WEB
100026	Branch Card Issuance Fee-WEB
100027	Card Order - Payment-WEB
100028	Card Replacement Request-WEB
100029	Card Replacement Fee-WEB
100030	CallCenter Replacement (Due to Damanged Card)-WEB
100031	CallCenter Replacement (Due to Damanged Card) Fee-WEB
100032	CallCenter Card Cancellation-WEB
100033	Customer Details-WEB
100034	Customer Refund Request-WEB
100035	SMS Balance Enquiry-WEB
100036	SMS Balance Enquiry Fee-WEB
100037	Call Center Request PAC-WEB
100038	CallCenter Replacement-Instant-WEB
100039	CallCenter Replacement-Instant-Fee-WEB
100041	Lost Card Blocking-WEB
100042	Customer Lost Card Blocking Fee-WEB
100043	Customer Refund Request Process-WEB
100044	Online Card Registration-WEB
100045	New Card Account-WEB
100046	New Card Account Fee-WEB
100048	New Card Account Subscription Fee-WEB
100049	Branch Refund Maker-WEB
100050	Branch Refund-WEB
100051	Branch Refund Fee-WEB
100052	Refund-WEB
100053	Refund Fee-WEB
100054	Topup-WEB
100055	TopUp Fee-WEB
100056	Customer Locking/Unlocking-WEB
100057	Online Purchase Fee-WEB

100058	Registered Post-WEB
100059	Registered Post Fee-WEB
100060	Online Purchase-WEB
100061	Council Registration-WEB
100062	Card Statement Request-WEB
100064	Card Statement Transaction Fee-WEB
100067	Web Access PIN-WEB
100068	International Transaction Fee-- Non - WEB [Cross Border Conversion Fee]
100069	Customer Dispute-WEB
100070	Customer Dispute Fee-WEB
100071	International Transaction Fee- Non - WEB [Cross Border Fee]
100072	International Transaction Fee- Non - WEB [Conversion Fee]
100073	Conversion Fee- Non - WEB [Not used in PPH]
100074	CallCenter Card Not Received Fee-WEB
100075	CallCenter Transactions-WEB
100076	CallCenter Audit Trail-WEB
100077	CallCenter Lost/Stolen Card-WEB
100078	CallCenter Transactions Fee-WEB
100079	CallCenter Notepad-WEB
100080	CallCenter Email-WEB
100081	CallCenter Damaged Card-WEB
100082	CallCenter Customer Details-WEB
100083	CallCenter Found Card-WEB
100084	CallCenter Found Card Fee-WEB
100085	CallCenter Damaged Card Fee-WEB
100086	Customer Card Replacement Request Fee-WEB
100087	Customer Account Summary-WEB
100089	Transaction History-WEB
100090	Dispute Logging-WEB
100091	Update My Details-WEB
100092	Change Password-WEB
100093	Call Center Card History-WEB
100094	Card Reissue Request-WEB
100095	Entity Payment-WEB
100098	Customer - New Card Account-WEB
100099	Customer - New Card Account Fee-WEB
100100	Product Settings-WEB
100101	WEB Reset PIN-WEB

100102	WEB Reset PIN Fee-WEB
100103	Monthly Fee Transaction-Non - WEB
100104	Monthly Fee- Non - WEB
100105	Deactivation through Merchant- Non - WEB
100106	Deactivation through Merchant- Non - WEB
100116	Branch TopUp-WEB
100117	Branch TopUp-WEB
100118	Branch TopUp Fee-WEB
100119	Fund Transfer Debit Transaction-WEB
100120	Fund Transfer Credit Transaction-WEB
100123	Activation through Merchant Fee Non - WEB
100124	Activation through Merchant Fee Non - WEB
100125	Refund Reversed Non - WEB
100128	Customer Card Replacement Request
100129	Customer Card Replacement Fee
100130	Supplimentary Card
100131	Supplimentary Card Fee
100132	Account Authorization
100134	Branch TopUp Maker
100135	Account Delink Authorization
100201	Call Center Change PIN IMG
100203	Call Center New Card Account IMG
100204	Call Center Fund Transfer IMG
100205	Call Center Auth Order Settings IMG - WEB
100206	Call Center Auth Order Settings View IMG - WEB
100207	Call Center Lost / Stolen Card IMG-WEB
100208	Call Center Card Not Received IMG-WEB
100209	Call Center Transactions IMG-WEB
100210	Call Center Request PAC IMG-WEB
100211	Call Center Card Activation IMG-WEB
100212	Call Center Lost / Stolen / Damaged Card IMG-WEB
100213	Call Center Change PAC IMG-WEB
100239	FeeRuler Filter Adjustment-Credit-WEB
100240	FeeRuler Filter Adjustment Fee-Credit-WEB
100241	FeeRuler Filter Adjustment-Debit-WEB
100242	FeeRuler Filter Adjustment Fee-Debit-WEB
100243	CallCenter Card Replacement-WEB
100244	CallCenter Card Replacement Fee-WEB
100245	CallCenter Customer Balance-WEB
100246	CallCenter Customer Balance Fee-WEB

100247	CallCenter Locking/Unlocking-WEB
100248	CallCenter Locking/Unlocking Fee-WEB
100249	CallCenter Lost/Stolen/Damaged Card-WEB
100250	CallCenter Lost/Stolen/Damaged Card Fee-WEB
100261	Online Card Issuance-WEB
100262	Online Card Issuance Fee-WEB
100290	Purchaser Registration-WEB
100292	Net Card Issuance-WEB
100293	Net Card Issuance Fee-WEB
100294	NET New Card Account-WEB
100295	NET New Card Account Fee-WEB
100296	Branch Card issuance-WEB
100297	Branch Card issuance Fee-WEB
100328	Customer Card Replacement Request1-WEB
100329	Customer Card Replacement Fee1-WEB
100555	Customer Locking/Unlocking Fee-WEB
100901	Call Center Customer Balance Fee-WEB
100903	Agency Match Off-WEB
100911	Block Swap Amount-Non - WEB
100912	Swap Transfer Debit Transaction- Non - WEB
100913	Swap Transfer Debit Transaction Fee - Non - WEB
100914	Swap Transfer Credit Transaction - Non - WEB
100915	Swap Transfer Credit Transaction Fee- Non - WEB
100916	Release Swap Amount- Non - WEB
101011	Chargeback Details- WEB & NON-WEB
101012	Outstanding- WEB & NON - WEB
101025	Technical Chargeback- WEB
101100	Charge Fees-WEB
101150	Fee Collection-Non - WEB
101160	Fee Disbursement - Non - WEB
101170	Fee Collection- Non - WEB
101180	Fee Disbursement- Non - WEB
108042	Chargeback Request-WEB
109001	Fund Transfer Third party-WEB
109002	Fund Transfer Third party Fee-WEB
110001	Peso Fund Transfer with in EastWest Bank
110002	Peso Fund Transfer with in EastWest Bank Fee
110003	Peso Fund Transfer with in EastWest Bank Same customer
110004	Peso Fund Transfer with in EastWest Bank Same

	customer Fee
110005	Dollar Fund Transfer with in EastWest Bank
110006	Dollar Fund Transfer with in EastWest Bank Fee
110007	Dollar Fund Transfer with in EastWest Bank Same customer
110008	Dollar Fund Transfer with in EastWest Bank Same customer Fee
110009	Peso Fund Transfer Intra Bank
110010	Peso Fund Transfer Intra Bank Fee
110011	Dollar Fund Transfer Intra Bank
110012	Dollar Fund Transfer Intra Bank Fee
110013	International Fund Transfer-Non - WEB
110014	International Fund Transfer Fee- Non - WEB
110015	Bills Payment Debit
110016	Bills Payment Debit Fee
110017	Bills Payment Credit
110018	Bills Payment Credit Fee
110019	Cheque Book Re-Order
110020	Cheque Book Re-Order Fee
110021	Statement of Account Fee for Debit
110022	Statement of Account Fee for Credit
110023	Online Cheque Viewing Fee
110024	Cheque Stop Payment Order Fee
110025	Statement Request Fee-WEB
110026	Lost Card Blocking Limit-WEB
110027	Lost Card Blocking Fee-WEB
110028	Card Locking Fee-WEB
110030	Online Foreign Exchange Fee
110031	Open a Savings Account
111111	Card Balance- Non - WEB
111112	SMS Alert- Non - WEB
111113	SMS Alert Fee- Non - WEB
200000	Refund- Non - WEB
200003	Balance Inquiry- Non - WEB [Not used in PPH]
200005	POS Cash Advance- Non - WEB
200006	Cash Advance- Non - WEB [Domestic Cross Currency based on Billing ccy for authorization]
200007	Cash Advance- Non - WEB [International Cross currency based on Billing ccy for authorization]
200008	Cash Advance Fee- Non - WEB
200009	Cash Advance- Non - WEB[Domestic Cross Currency]

200074	Forex- Non - WEB
200079	Forex- Non - WEB
201000	Merchant Credit Return
210000	IBFT Deposit
211128	Financial Adjustment- Non - WEB
230007	Cash Advance- Non - WEB [SMS Cash Advance]
230008	Cash Advance- Non - WEB [SMS Domestic Cross Currency based on Billing ccy authorization]
230009	Cash Advance- Non - WEB [SMS International Cross currency based on Billing ccy authorization]
230010	Cash Advance- Non - WEB [Domestic Cross Currency]
250000	Language Change-WEB
250001	LangChange Fee-WEB
260000	VMT amount
260001	VMT Fee
270000	Account Verification Service
280000	Cash Payment
290000	Funds Disbursement Credit
300000	POS Balance Inquiry-- Non - WEB
300001	Balance Inquiry Fee- Non - WEB
300006	Sale Fee- Non - WEB
300007	Cash Back- Non - WEB
300008	Cash Back Fee- Non - WEB
300020	Decline POS Balance Inquiry- Non - WEB
300021	Decline POS Balance Inquiry Fee- Non - WEB
302551	Balance Adjustment Debit- Non - WEB
302552	Balance Adjustment Credit- Non - WEB
310000	ATM Balance Inquiry- Non - WEB
310001	ATM Balance Inquiry Fee - Non - WEB
310020	Decline ATM Balance Inquiry- Non - WEB
310021	Decline ATM Balance Inquiry Fee- Non - WEB
320000	Oar
320001	Oar Fee
322239	Card Adjustment-WEB
400000	ATM Fund Transfer- Non - WEB
400001	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400002	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400003	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400004	Transaction Settlement- Non - WEB [Used in Dispute for

	FA]
400005	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400006	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400007	Transaction Settlement- Non - WEB [Used in Dispute for FA]
400008	Copy Request-WEB [Used in Dispute for FA]
400010	ATM Fund Transfer Credit Transaction- Non - WEB [Not Used in PPH]
400011	ATM Fund Transfer Fee- Non - WEB [Not Used in PPH]
402021	FUND TRANSFER INTERBANK Credit Transaction
440000	FUND TRANSFER INTER
460000	IBFT WITHDRAWAL
460001	IBFT Withdrawal Fee
500001	Purchase Reversal- Non - WEB[Domestic Same Currency]
500002	Purchase Reversal- Non - WEB[Domestic Cross Currency]
500003	Purchase Reversal- Non - WEB[International Same Currency]
500004	Purchase Reversal- Non - WEB[International Cross Currency]
500005	Cash Withdrawal Reversal- Non - WEB[Domestic Same Currency]
500006	Cash Withdrawal Reversal- Non - WEB[Domestic Cross Currency]
500007	Cash Withdrawal Reversal- Non - WEB[International Same Currency]
500008	Cash Withdrawal Reversal- Non - WEB[International Cross Currency]
500009	Cash Disbursement Reversal- Non - WEB
500010	IBFT Withdrawal
500011	Bill Payment- Non - WEB
500012	Purchase Release- Non - WEB
500013	Cash Withdrawal Reversal- Non - WEB
500014	Purchase Reversal- Non - WEB[SMS Domestic Same Currency]
500015	Cash Disbursement Reversal- Non - WEB[SMS Cash Disbursement]
500016	Cash Disbursement Reversal- Non - WEB
500017	Merch Return Reversal- Non - WEB
500018	Cash Withdrawal Reversal- Non - WEB[Domestic Cross Currency based on Billing ccy authorization]
500019	Cash Withdrawal Reversal- Non - WEB[International Cross currency based on Billing ccy authorization]
500020	Purchase Reversal- Non - WEB[Domestic Cross Currency]

	based on Billing ccy authorization]
500021	Purchase Reversal- Non - WEB[International Cross currency based on Billing ccy authorization]
500022	Cash Disbursement Reversal- Non - WEB[SMS Domestic Cross Currency based on Billing ccy authorization]
500023	Cash Disbursement Reversal- Non - WEB[SMS International Cross currency based on Billing ccy authorization]
500024	Cash Disbursement Reversal- Non - WEB[SMS Domestic Cross Currency]
500025	Cash Disbursement Reversal- Non - WEB[Domestic Cross Currency]
500026	Purchase Release- Non - WEB
500027	Merchant Return Reversal- Non - WEB
500028	Cash Advance Release- Non - WEB[Domestic Cross Currency based on Billing ccy authorization]
500029	Cash Advance Release- Non - WEB[International Cross currency based on Billing ccy authorization]
500030	Cash Advance Release- Non - WEB[Domestic Same Currency]
500031	Cash Advance Release- Non - WEB[Domestic Cross Currency]
500032	Cash Withdrawal Reversal- Non - WEB[Domestic Cross Currency]
500033	Purchase Reversal- Non - WEB[Domestic Cross Currency]
500034	Cash Advance Reversal- Non - WEB[Domestic Cross Currency]
500035	Cash Withdrawal Reversal- Non - WEB[Domestic Same Currency]
500036	Purchase Reversal- Non - WEB[Domestic Same Currency]
500037	Cash Advance Reversal- Non - WEB[Domestic Same Currency]
500038	Cash Withdrawal Reversal- Non - WEB[International Same Currency]
500039	Purchase Reversal- Non - WEB[International Same Currency]
500040	Cash Withdrawal Reversal- Non - WEB[International Cross Currency]
500041	Purchase Reversal- Non - WEB[International Cross Currency]
500074	Forex Reversal
500075	Forex Reversal
500076	Forex Reversal
570006	Cash Withdrawal Reversal- Non - WEB[SMS Domestic Cross Currency]
570007	Cash Withdrawal Reversal- Non - WEB[SMS International

	Same Currency]
570008	Cash Withdrawal Reversal- Non - WEB[SMS International Cross Currency]
580002	Purchase Reversal- Non - WEB[SMS Domestic Cross Currency]
580003	Purchase Reversal- Non - WEB[SMS International Same Currency]
580004	Purchase Reversal- Non - WEB[SMS International Cross Currency]
600000	Transaction Settlement [Used in EOD and Dispute for FA entries]
600001	Chargeback-WEB[Used in EOD and Dispute for FA entries]
600002	Transaction Settlement- Non - WEB[Used in EOD and Dispute for FA entries]
600003	Chargeback Reconciliation-WEB[Used in EOD and Dispute for FA entries]
600004	Chargeback- Non - WEB & WEB[Used in EOD and Dispute for FA entries]
600005	Chargeback- Non - WEB & WEB[Used in EOD and Dispute for FA entries]
600008	CustomerCreditQueue-WEB[Used in EOD and Dispute for FA entries]
600009	Transaction Settlement - Non - WEB[Used in EOD and Dispute for FA entries]
600011	Chargeback Representment-WEB[Used in Dispute for FA entries]
600012	Technical Chargeback-WEB[Used in Dispute for FA entries]
600013	Chargeback Representment-WEB[Used in Dispute for FA entries]
700000	Manual Load/UnLoad-WEB
700001	Balance Transfer-WEB
700002	Fee Reversal-WEB
700003	Reversal Chargeback-[Used in Dispute for FA entries]
700004	Close Account
700005	Fraud
700006	Service Enquiry Fee
700007	Dispute- WEB[Used in Dispute for FA entries]
700008	Other-WEB[Used in Dispute for FA entries]
800000	Refund - Non WEB[Used in Dispute for FA entries]
800001	Refund Settlement - Non WEB[Used in Dispute for FA entries]
800002	Release-WEB[Used in Dispute for FA entries]
800003	Reversal Settlement-Non WEB[Used in Dispute for FA entries]

800004	Refund Reversal Settlement-Non WEB[Used in Dispute for FA entries]
800005	Outstanding-WEB[Used in EOD for FA entries]
800006	Refund-WEB[Used in Dispute for FA entries]
800008	Outstanding-WEB[Used in Dispute for FA entries]
800009	Refund Settlement[Used in EOD for FA entries]
800010	Refund[Used in EOD for FA entries]
800011	Reversal Settlement[Used in Dispute for FA entries]
800012	Representment- WEB[Used in Dispute for FA entries]
800013	Chargeback Representment-WEB[Used in Dispute]
800014	Chargeback Representment-WEB[Used in Dispute]
800015	Chargeback Representment-WEB[Used in Dispute]
800016	Chargeback Representment-WEB[Used in Dispute]
800017	Chargeback Representment-WEB[Used in Dispute]
800018	Chargeback Representment-WEB[Used in Dispute]
800020	Chargeback Reconciliation-WEB[Used in Dispute]
800021	Reissue Fee-WEB[Used in Dispute]
800022	Maintenance Fee-WEB[Used in Dispute]
840000	BILL PAYMENT- WEB
900100	IVR Balance Inquiry- Non WEB
900101	IVR Balance Inquiry Fee Non WEB
900102	IVR Last n Transactions Non WEB
900103	IVR Last n Transactions Fee - Non WEB
900104	IVR Enquiry- Non WEB
900105	IVR Enquiry Fee-Non - WEB
900106	IVR Reset PIN- Non - WEB
900107	IVR Reset PIN Fee - Non - WEB
900108	Council Maker-WEB
900110	Reorder Maker-WEB
900111	Fee Maker-WEB
900112	Assign Products-WEB
900113	Card Activation-WEB
900114	Pin Regeneration-WEB
900115	Card Replacement Authorization-WEB
900116	Service User Registration-WEB
900117	Service User Registration Fee-WEB
910000	Check Book Reorder-WEB
940000	Pin Change-Non - WEB
940001	Pin Change Fee Non - WEB
950000	Mini Statement-WEB

950001	MST Fee-WEB
950002	Business Client Top up Debit FileUpload-WEB
950003	Business Client Top up Credit FileUpload-WEB
950004	Business Client/Consumer Top up-WEB
950005	Business Client Top up Debit WEB
950006	Business Client Top up Credit WEB
950007	Business Client Top up Fee FileUpload
950008	Business Client Top up Fee WEB
950009	Consumer Top up Fee
950010	Business Client Refund Debit WEB
950011	Business Client Refund Fee WEB
950012	Business Client Refund Credit WEB
950013	Business Client Registration- WEB
950014	Consumer Registration-WEB
950016	Basic Due Diligence-WEB
950017	Customer FundTransfer Debit Transaction-WEB
950018	Customer FundTransfer Credit Transaction-WEB
950019	Customer FundTransfer Fee-WEB
950020	Customer Card Closure Refund-WEB
950021	Customer Card Closure Refund Fee-WEB
950022	Customer Card Closure Refund Unsettle-WEB
950023	Customer Card Closure Refund Unsettle Fee-WEB
950024	Microsite Customer Topup-WEB
950025	Microsite Customer Topup Fee-WEB
950026	Product Upgrade-WEB
950027	Council Refund Debit FileUpload-WEB
950028	Council Refund Credit FileUpload-WEB
950029	Council Refund Fee FileUpload-WEB
950031	Council/Service User Refund-WEB
950032	Service User Refund Fee-WEB
CRPFUP	Council Registration - File Upload-WEB
CRPTOP	Reload - Manual-WEB
EMPFUP	Employee Registration - File Upload-WEB
EPAYPC	Entity Payment- WEB

7.5 List of Transaction Types

These are generalized transaction types for common API specification. It is changed through client based.

Transaction Type	Meaning
T	Approved [Web Transactions]
T	Pending
F	Pending [Fee Record]
R	Reversed
RF	Fee Reversed
S	Settled
TR	Transaction Reversed
TF	Fee Reversed
BR	Reversed [Base2 Reversal]
FBR	Fee Reversed [Base2 Reversal record Fee]
RL	Released
FRL	Fee Released
CB	Chargeback Raised
DC	Dispute Closed
CR	Copy Request Raised
DP	ChargeBack Pending